
Floating-Point Networks with Automatic Differentiation

Can Represent Almost All Floating-Point Functions and Their Gradients

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Abstract

Theoretical studies show that for any differentiable function on a compact domain, there exists a neural network that approximates both the function values and gradients. However, such results do not directly apply to practical models since they assume real parameters and exact internal operations. In contrast, real implementations only use a finite subset of reals and machine operations with round-off errors. In this work, we investigate whether a similar result holds for neural networks under floating-point arithmetic, when the gradient with respect to the input is computed by the automatic differentiation algorithm D^{AD} . We first show that given a floating-point function ϕ (e.g., a loss function), arbitrary function values and gradients can be represented by a floating-point network f and $D^{\text{AD}}(\phi \circ f)$, respectively. We further extend this result: given $\phi_1, \dots, \phi_n$, $D^{\text{AD}}(\phi_i \circ f)$ can simultaneously represent arbitrary gradients while f represents the target values, under mild conditions. Our results hold for practical activation functions, e.g., ReLU, ELU, GELU, Swish, Sigmoid, and tanh.

1. Introduction

Modern neural network applications often require not only the function values but also the gradients with respect to the input. For example, in the field of scientific machine learning, neural networks are employed as surrogates for physical models or solutions to partial differential equations (Raissi et al., 2019). Gradients are also used in various problems, such as sensitivity analysis (Simonyan et al., 2014), inverse

problems (Bora et al., 2017), and reinforcement learning (Lillicrap et al., 2016), where decisions depend on how outputs change in response to small perturbations of inputs. Furthermore, gradients are often used to revise inputs, e.g., for adversarial attacks (Goodfellow et al., 2015; Fredrikson et al., 2015) and generation (Gatys et al., 2016). They are also used to improve the performance of trained networks, e.g., several studies show that fitting the gradient together with the function value can increase the final performance of networks (Czarnecki et al., 2017; Gulrajani et al., 2017).

Theoretical results show that neural networks can approximate complex functions when the networks use real parameters and exact operations (e.g., addition and multiplication). Early results focused on fully-connected networks (Pinkus, 1999), which have been extended to modern architectures such as convolutional neural networks (Zhou, 2020), recurrent neural networks (Schäfer & Zimmermann, 2007), residual networks (Lin & Jegelka, 2018), and transformers (Yun et al., 2020). While most of these results focus on approximating function values only, it is also known that neural networks can approximate function values and gradients at the same time (Li, 1996; Hwang, 2025).

In practice, neural networks are executed on computers using floating-point arithmetic (IEEE, 2019). All network parameters and intermediate values are represented by a finite set of floating-point numbers, and internal operations follow the rules of floating-point computations, involving round-off errors. Hence, a practical network is an algorithm determined by a sequence of floating-point operations rather than a real-valued function in theory. Likewise, gradients in such systems are computed by applying chain-rule-based automatic differentiation algorithms (e.g., backpropagation) to this computational graph (Griewank & Walther, 2008). Therefore, this gradient is not the derivative of a differentiable, real-valued function, but rather the output of a computational procedure.

A few recent studies have investigated the representability of floating-point neural networks that only use floating-point parameters and operations. Park et al. (2024) show that floating-point networks can represent all functions from floating-point vectors to floating-point values when the activation function is ReLU or the binary step function. This

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result has been extended to general activation functions (Hwang et al., 2025b) and floating-point interval arithmetic (Hwang et al., 2025a). Nevertheless, as in most existing results under exact arithmetic, these results aim to fit the function values only, not the gradients.

1.1. Contributions

In this work, we study the representability of floating-point networks together with automatic differentiation (AD). Specifically, we investigate the following problem:

Can floating-point networks with AD represent arbitrary function values and arbitrary gradients?

To better describe our results, we briefly introduce a floating-point network f consisting of floating-point affine transformations $\rho_1, \dots, \rho_l$ and a pointwise activation function σ :

$$f(x) = \rho_l \circ \sigma \circ \rho_{l-1} \circ \sigma \circ \dots \circ \sigma \circ \rho_1(x).$$

For a floating-point function ϕ_x (e.g., a loss function that may depend on the input x) AD for $\phi_x \circ f$ is computed as follows: for predefined derivatives $\phi'_x, \rho'_i, \sigma'$ of ϕ_x, ρ_i, σ ,

$$D^{\text{AD}}(\phi_x \circ f, x) = \phi'_x(f(x)) \otimes \rho'_l(z_l) \otimes \sigma'(y_{l-1}) \otimes \dots \otimes \rho'_1(x)$$

where z_i denotes the input to ρ_i , y_i denotes the input to σ in the layer i , and $\otimes$ denotes the floating-point multiplication. Importantly, the multiplications here should be computed from left to right; otherwise, the final value can be different due to the non-associativity of floating-point operations.

In our first result sketched below, we prove that floating-point networks with AD can represent almost all function values and gradients, where $\mathbb{F}$ denotes the set of finite floating-point numbers.

Theorem 3.1 (informal). *For almost all $f^* : \mathbb{F}^d \rightarrow \mathbb{F}^d$, $g^* : \mathbb{F}^d \rightarrow \mathbb{F}^d$, and $\phi'_x : \mathbb{F} \rightarrow \mathbb{F}$ such that $g^*(x) = 0$ for all x with $\phi'_x(f^*(x)) = 0$, there exists a floating-point network f such that $f = f^*$ and $D^{\text{AD}}(\phi_x \circ f, x) = g^*(x)$.*

Theorem 3.1 states that given some function ϕ'_x , f and $D^{\text{AD}}(\phi_x \circ f)$ can represent the target function f^* and the gradient g^* , respectively. Here, we do not consider the antiderivative ϕ_x of ϕ'_x since it does not affect $D^{\text{AD}}(\phi_x \circ f)(x)$. Furthermore, ϕ'_x can be arbitrarily chosen. For example, if $\phi'_x(x) = 1$ for all x , then $D^{\text{AD}}(\phi_x \circ f)$ corresponds to the gradient of f computed by AD. For general ϕ'_x , Theorem 3.1 shows that floating-point networks can represent the gradient of $\phi_x \circ f$, as long as the value of ϕ'_x is non-zero.

Theorem 3.1 implies that floating-point networks can solve problems that require fitting both function values and gradients simultaneously, e.g., (Raissi et al., 2019; Czarnecki et al., 2017). Furthermore, Theorem 3.1 shows that floating-point networks can successfully perform target tasks (by

fitting function values) while having manipulated gradients (e.g., zero for all inputs). Such manipulation can be useful when one does not want to leak information from gradients, e.g., to protect against gradient-based attacks and reasoning (Goodfellow et al., 2015; Simonyan et al., 2014).

Theorem 3.1 applies to various practical activation functions, including ReLU, ELU, GELU, Swish, Sigmoid, and tanh. Here, the activation function determines the possible range of g^* and ϕ'_x . For example, for ReLU, ELU, GELU, and Swish, Theorem 3.1 covers all $f^*, g^* : [-\Omega/8, \Omega/8]^d \rightarrow \mathbb{F}$ and $\phi'_x : \mathbb{F} \rightarrow \mathbb{F}$ where Ω denotes the largest finite float, e.g., $\Omega \approx 2^{128}$ and 2^{16} for the 32-bit single-precision and 16-bit half-precision formats (IEEE, 2019). For Sigmoid, Theorem 3.1 covers $f^*, g^* : [-\Omega/4, \Omega/4]^d \rightarrow \mathbb{F}$ and $\phi'_x : \mathbb{F} \rightarrow [-\Omega/4, \Omega/4]$, while tanh covers a slightly smaller range of ϕ'_x .

The floating-point network f in Theorem 3.1 is a function of ϕ'_x . Hence, given f , $D^{\text{AD}}(\phi_x \circ f, x)$ is a function of x only (i.e., $g^*(x)$), and it may output an arbitrary gradient for a different choice of ϕ'_x (e.g., when ϕ_x attached to f changes). Then, can we construct a floating-point network f so that $D^{\text{AD}}(\phi_x \circ f, x)$ is a function of ϕ'_x and x given f ? We provide a positive answer in our next result.

Theorem 3.2 (informal). *For almost all $f^* : \mathbb{F}^d \rightarrow \mathbb{F}^d$ and $g^* : \mathbb{F}^d \times \mathbb{F} \rightarrow \mathbb{F}^d$ such that $g^*(x, -y) = -g^*(x, y)$, there exists a floating-point network f such that for almost all $\phi'_x : \mathbb{F} \rightarrow \mathbb{F}$, $f = f^*$ and $D^{\text{AD}}(\phi_x \circ f)(x) = g^*(x, \phi'_x(f(x)))$.*

Theorem 3.2 implies that there exists a single floating-point network f such that, even when ϕ'_x changes, $D^{\text{AD}}(\phi_x \circ f, x)$ changes accordingly, following $g^*(x, \phi'_x(f(x)))$. Importantly, we can choose an arbitrary $g^*(x, \phi'_x(f(x)))$ as long as $g^*(x, -y) = -g^*(x, y)$; such a restriction cannot be bypassed due to the symmetry inherent in the definition of $D^{\text{AD}}(\phi_x \circ f, x)$. We note that an analogous result cannot hold under exact arithmetic since

$$\nabla(\tilde{\phi} \circ \tilde{f})(x) = \tilde{\phi}'(\tilde{f}(x)) \times \nabla f(x)$$

should be proportional to $\tilde{\phi}'(\tilde{f}(x))$ for real differentiable functions $\tilde{\phi}$ and $\tilde{f}$ under exact arithmetic. Nevertheless, under floating-point arithmetic, it is possible to represent gradients that are not proportional to ϕ'_x by exploiting the non-associativity of floating-point multiplication. That is, floating-point networks can perform the target task by fitting function values while having artificially manipulated gradients, even if ϕ'_x can change over time. We note that Theorem 3.2 also applies to floating-point networks using activation functions that we discussed after Theorem 3.1. However, our results are not limited to these activation functions; we also provide similar results for general activation functions in Theorems 3.6 and 3.8.

1.2. Organization

We introduce the notation and precise problem setups in Section 2. We then formally present our main results and describe their proof outlines in Section 3. The formal proofs of our main results are given in Section 4. We finally conclude the paper in Section 5.

2. Preliminaries

2.1. Notation

Throughout this paper, $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{R}$ denote the sets of natural numbers, integers, and real numbers, respectively. We also define $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$. For $x \in \mathbb{R}$, the ceiling function is defined by $\lceil x \rceil_{\mathbb{Z}} := \min\{z \in \mathbb{Z} \mid z \geq x\}$. For $a, b \in \mathbb{R}$ and a set $\mathcal{S}$, we define $[a, b]_{\mathcal{S}} := [a, b] \cap \mathcal{S}$, where $[a, b] \subset \mathbb{R}$ denotes the closed interval. We also use $(a, b)_{\mathcal{S}}$, $(a, b]_{\mathcal{S}}$, and $[a, b)_{\mathcal{S}}$ analogously. For $d \in \mathbb{N}$, we use $\mathbf{0}_d := (0, \dots, 0) \in \mathbb{R}^d$, $\mathbf{1}_d := (1, \dots, 1) \in \mathbb{R}^d$ and $[d] := \{1, \dots, d\}$. For $k \in [d]$, let e_k denote the k -th standard basis vector in $\mathbb{R}^d$, whose k -th component is equal to one and the remaining components are zero. For $x \in \mathbb{R}^d$ and $k \in [d]$, x is always a column vector, and x_k denotes the k -th component of x ; that is, $x = (x_1, \dots, x_d)$. The same notation applies to vector-valued functions f with codomain $\mathbb{R}^d$, i.e., $f = (f_1, \dots, f_d)$. For a function $f : \mathcal{X} \rightarrow \mathcal{X}$ and $n \in \mathbb{N}$, the n -fold composition of f is denoted by f^{on} . For a set $\mathcal{X}$, $d_1, d_2 \in \mathbb{N}$, $x_1 = (x_{11}, \dots, x_{1d_1}) \in \mathcal{X}^{d_1}$, and $x_2 = (x_{21}, \dots, x_{2d_2}) \in \mathcal{X}^{d_2}$, we define $x_1 || x_2 := (x_{11}, \dots, x_{1d_1}, x_{21}, \dots, x_{2d_2}) \in \mathcal{X}^{d_1+d_2}$. We also use $\left| \left| \right|_{i=1}^n x_i := ((\dots (x_1 || x_2) || x_3) \dots || x_n)$.

Let $\mathcal{X}, \mathcal{Y} \subset \mathbb{F}^{d_1}$ and $\alpha, \beta \in \mathbb{F}^{d_2}$. We say “a function $f : \mathbb{F}^{d_1} \rightarrow \mathbb{F}^{d_2}$ maps $(\mathcal{X}, \mathcal{Y})$ to (α, β) ” if

$$f(x) = \begin{cases} \alpha & \text{if } x \in \mathcal{X}, \\ \beta & \text{if } x \in \mathcal{Y}. \end{cases}$$

We denote this by $f : (\mathcal{X}, \mathcal{Y}) \mapsto (\alpha, \beta)$. For a singleton set $\mathcal{X} = \{x_0\}$ and for $\mathcal{Y} = \{y_0\}$ or $\mathbb{F}^{d_1} \setminus \{x_0\}$, we omit the set braces and $\mathbb{F}^{d_1}$ and simply write $f : (x_0, y_0) \mapsto (\alpha, \beta)$ or $f : (x_0, \setminus x_0) \mapsto (\alpha, \beta)$. If the value of $\mathcal{Y}$ is not important, we simply write $f : \mathcal{X} \mapsto \alpha$.

2.2. Floating-Point Arithmetic

In this paper, we follow the IEEE-754 standard for floating-point arithmetic (IEEE, 2019). The set of finite floating-point numbers $\mathbb{F}_{M,E} \subset \mathbb{R}$ is defined as

$$\mathbb{F}_{M,E} := \{s \times (1 + i/2^M) \times 2^e, s \times (i/2^M) \times 2^{\epsilon_{\min}} \mid s \in \{-1, 1\}, i \in [0, 2^M - 1]_{\mathbb{Z}}, e \in [\epsilon_{\min}, \epsilon_{\max}]_{\mathbb{Z}}\}$$

where $\epsilon_{\min} := -2^{E-1} + 2$ and $\epsilon_{\max} := 2^{E-1} - 1$. When it is clear from the context, we omit M, E , and use $\mathbb{F} := \mathbb{F}_{M,E}$.

We say that “ $x \in \mathbb{F}$ is even” if the i in the representation of x (i.e., $s(1 + i/2^M)2^e$ or $s(i/2^M)2^{\epsilon_{\min}}$) is even. The set of all floating-point numbers $\overline{\mathbb{F}}$ contains some non-finite numbers: $\overline{\mathbb{F}} := \mathbb{F} \cup \{-\infty, \infty, \text{NaN}\}$. However, we note that non-finite floats $-\infty, \infty$, and NaN will not appear during the intermediate computations of our network constructions, and hence, we will mainly focus on operations on $\mathbb{F}$. For $x \in \mathbb{F}$, we define $x^+ := \min\{y \in \overline{\mathbb{F}} \mid y > x\}$ and $x^- := \max\{y \in \overline{\mathbb{F}} \mid y < x\}$. The largest and smallest positive finite floating-point numbers are denoted by $\Omega := (2 - 2^{-M})2^{\epsilon_{\max}}$ and $\omega := 2^{\epsilon_{\min} - M}$, respectively, e.g., $\Omega^+ = \infty$.

To define floating-point operations, we define the floating-point rounding operation $\lceil \cdot \rceil : \mathbb{R} \cup \{-\infty, \infty, \text{NaN}\} \rightarrow \mathbb{F}$ as follows:

$$\lceil x \rceil := \begin{cases} \arg \min_{y \in \mathbb{F}} |x - y| & \text{if } |x| < \Omega + 2^{\epsilon_{\max} - M - 1}, \\ -\infty & \text{if } x \leq -\Omega - 2^{\epsilon_{\max} - M - 1}, \\ \infty & \text{if } x \geq \Omega + 2^{\epsilon_{\max} - M - 1}, \\ \text{NaN} & \text{if } x = \text{NaN}. \end{cases}$$

If a tie occurs in $\arg \min_{y \in \mathbb{F}} |x - y|$, we choose the even y . For $\hat{\sigma} : \mathbb{R} \rightarrow \mathbb{R}$, we define its rounded version $\lceil \hat{\sigma} \rceil : \overline{\mathbb{F}} \rightarrow \overline{\mathbb{F}}$ as $\lceil \hat{\sigma} \rceil(x) := \lceil \hat{\sigma}(x) \rceil$ for $x \in \mathbb{F}$; see Appendix A.2 for the definition of $\lceil \hat{\sigma} \rceil$ on non-finite floats. While the rounding operation can output infinities and NaN, we note that all the outputs of the rounding operations in this paper will be finite. For the real operations $+$, $-$, and $\times$, we define their floating-point counterparts as follows: for $x, y \in \mathbb{F}$, $x \oplus y := \lceil x + y \rceil$, $x \ominus y := \lceil x - y \rceil$, and $x \otimes y := \lceil x \times y \rceil$. The definition of floating-point operations when x or y is in $\{-\infty, \infty, \text{NaN}\}$ is deferred to Appendix A.2.

Since floating-point operations are not associative in general, the order of evaluation must be specified. For $n \in \mathbb{N}$ and $x_1, \dots, x_n \in \overline{\mathbb{F}}$, we define $\bigoplus_{i=1}^n x_i := (\bigoplus_{i=1}^{n-1} x_i) \oplus x_n$ with the convention $\bigoplus_{i=1}^0 x_i := 0$. Unless otherwise specified by parentheses, summations are evaluated in a left-associative way, i.e., from left to right. In particular, we also compute $\bigoplus$ and $\bigotimes$ in a left-associative way: for $n_1, n_2 \in \mathbb{N}$ and $x_i, y_j \in \overline{\mathbb{F}}$,

$$\bigoplus_{i=1}^{n_1} x_i \oplus \bigoplus_{j=1}^{n_2} y_j := \left(\bigoplus_{i=1}^{n_1} x_i \oplus \bigoplus_{j=1}^{n_2-1} y_j \right) \oplus y_{n_2}.$$

The same ordering convention applies to $\bigotimes_{i=1}^{n_1} \bigotimes_{j=1}^{n_2} x_{i,j}$ for $x_{i,j} \in \overline{\mathbb{F}}$. When the output of the floating-point summations is not affected by the order, we may use $\bigoplus_{s \in \mathcal{S}} x_s$ for an index set $\mathcal{S}$ and $x_s \in \overline{\mathbb{F}}$.

We also extend floating-point multiplication to matrices, vectors, and scalars. For $x \in \mathbb{F}$ and $y \in \mathbb{F}^d$, we define $x \otimes y = y \otimes x := (x \otimes y_1, \dots, x \otimes y_d)$. Likewise, for $d_1, d_2 \in \mathbb{N}$, $A = [a_{i,j}]_{i,j} \in \mathbb{F}^{d_1 \times d_2}$, and $x \in \mathbb{F}^{d_2}$, we

define $A \otimes x$ as follows:

$$A \otimes x := \left(\bigoplus_{j=1}^{d_2} (a_{1,j} \otimes x_j), \dots, \bigoplus_{j=1}^{d_2} (a_{d_1,j} \otimes x_j) \right).$$

As in exact addition and multiplication, the operation $\otimes$ always takes precedence over $\oplus$.

Throughout this paper, we assume the following conditions on M , E , and σ' . Our assumption on M , E covers float16, float32, float64, bfloat16, 8bit-E5M2, and 8bit-E4M3 (see Table 2 in Appendix A).

Assumption 1. $2 \leq M \leq 2^{E-2}$ and $E \geq 4$.

2.3. Floating-Point Neural Network

We consider an activation function $\sigma : \mathbb{F} \rightarrow \mathbb{F}$ and its derivative $\sigma' : \mathbb{F} \rightarrow \mathbb{F}$. One may regard σ and σ' as the rounded versions of a real activation function and its derivative. Nevertheless, we do not impose any restrictions on σ, σ' and consider them as general functions from $\mathbb{F}$ to $\mathbb{F}$. We often apply σ and σ' to vectors: for $x = (x_1, \dots, x_d) \in \mathbb{F}^d$, $\sigma(x) = (\sigma(x_1), \dots, \sigma(x_d))$, and $\sigma'(x)$ is the $d \times d$ diagonal matrix whose i -th diagonal entry is $\sigma'(x_i)$. Throughout this paper, we assume that $\sigma'(\mathbb{F}) \subset \mathbb{F}$.

Let $L \in \mathbb{N}$ be the number of layers and $d_0, \dots, d_L \in \mathbb{N}$ denote the input/hidden/output dimensions. For each $l \in [L]$, let $\rho_l : \mathbb{F}^{d_{l-1}} \rightarrow \mathbb{F}^{d_l}$ be the (floating-point) affine transformation defined as $\rho_l(x) = A_l \otimes x \oplus b_l$ for some $A_l \in \mathbb{F}^{d_l \times d_{l-1}}$ and $b_l \in \mathbb{F}^{d_l}$. We define a floating-point neural network $f(\cdot; \theta) : \mathbb{F}^{d_0} \rightarrow \mathbb{F}^{d_L}$ with parameters $\theta = (A_1, \dots, A_L, b_1, \dots, b_L)$ as follows:

$$f(x; \theta) := \rho_L \circ \sigma \circ \rho_{L-1} \circ \dots \circ \sigma \circ \rho_2 \circ \sigma \circ \rho_1(x). \quad (1)$$

We call such an f “a σ network”. We often omit θ and use $f(x)$ to denote $f(x; \theta)$.

For convenience, we also define variants of floating-point networks. For f_1, f_2, f_3 defined as

$$\begin{aligned} f_1(x) &= \rho_L \circ \sigma \circ \dots \circ \sigma \circ \rho_1 \circ \sigma(x), \\ f_2(x) &= \sigma \circ \rho_L \circ \dots \circ \sigma \circ \rho_2 \circ \sigma \circ \rho_1(x), \\ f_3(x) &= \sigma \circ \rho_L \circ \dots \circ \sigma \circ \rho_1 \circ \sigma(x), \end{aligned}$$

we call f_1 “a network starting with σ ”, f_2 “a network ending with σ ”, and f_3 “a network starting and ending with σ ”. For these variants, we count the number of layers as the number of affine transformations they contain (i.e., all f_1, f_2 , and f_3 have L layers). For a (proper) network f defined in Eq. (1), one can observe that $f_1 \circ f \circ f_2$ is a floating-point network, and the number of layers in $f_1 \circ f \circ f_2$ is the sum of the numbers of layers in f_1, f_2, f .

For two L -layer networks $h_1(\cdot; \theta_1), h_2(\cdot; \theta_2) : \mathbb{F}^d \rightarrow \mathbb{F}$ with $\theta_i = (A_{i1}, \dots, A_{iL}, b_{i1}, \dots, b_{i(L-1)}, 0)$ for $i \in [2]$,

we define the operation $h_1 \# h_2$ by concatenating h_1 and h_2 as follows:

$$(h_1 \# h_2)(x; \tilde{\theta}) = \tilde{\rho}_L \circ \sigma \circ \dots \circ \sigma \circ \tilde{\rho}_1(x)$$

where $\tilde{\rho}_l(x) = \begin{bmatrix} A_{1l} & 0 \\ 0 & A_{2l} \end{bmatrix} \otimes x \oplus (b_{1l} \parallel b_{2l})$ for all $l \in [L-1] \setminus \{1\}$, $\tilde{\rho}_1(x) = (A_{11} \otimes x \oplus b_{11}, A_{21} \otimes x \oplus b_{21})$, $\tilde{\rho}_L(x) = [A_{1L} \ A_{2L}] \otimes x$ (note that A_{1L}, A_{2L} are row vectors), and $\tilde{\theta}$ is the corresponding parameters. Note that $h_1 \# h_2 : \mathbb{F}^d \rightarrow \mathbb{F}$ is also an L -layer network. We also define

$$\#_{i=1}^n f_i := ((\dots (f_1 \# f_2) \# f_3) \dots \# f_n).$$

2.4. Automatic Differentiation

Automatic differentiation (AD) computes derivatives by repeatedly applying local differentiation rules along a computational graph (Griewank & Walther, 2008). In modern deep learning frameworks such as PyTorch and TensorFlow, gradients are typically computed by reverse-mode AD (i.e., backpropagation), where gradients are propagated backward through the primitive operations used in the forward pass. Throughout this paper, we consider AD under floating-point arithmetic, meaning that both the forward computation and the backward differentiation procedure use floating-point operations with intermediate rounding.

To derive the gradients of floating-point networks computed by the (reverse-mode) automatic differentiation algorithm, we first define the derivatives of two types of *primitive functions*: affine transformations and activation functions. For $\rho : \mathbb{F}^{d_1} \rightarrow \mathbb{F}^{d_2}$ defined as $\rho(x) = A \otimes x \oplus b$ and for $\sigma : \mathbb{F}^{d_2} \rightarrow \mathbb{F}^{d_2}$, we define $\mathcal{D}_{\rho,x} : \mathbb{F}^{1 \times d_2} \rightarrow \mathbb{F}^{1 \times d_1}$ and $\mathcal{D}_{\sigma,x} : \mathbb{F}^{1 \times d_2} \rightarrow \mathbb{F}^{1 \times d_2}$ as follows: for $g \in \mathbb{F}^{1 \times d_2}$,

$$\mathcal{D}_{\rho,x}(g) := g \otimes A \text{ and } \mathcal{D}_{\sigma,x}(g) := g \otimes \sigma'(x).$$

Here, $\mathcal{D}_{\rho,x}(g)$ can be viewed as the gradient of $\phi_x \circ \rho$ for some $\phi_x : \mathbb{F}^{d_2} \rightarrow \mathbb{F}$, where $\phi'_x(\rho(x)) = g$. $\mathcal{D}_{\sigma,x}(g)$ can be considered in the same way.

These notations are useful when defining the gradient of a composite function: for $\psi = \psi_2 \circ \psi_1$, where ψ_2 and ψ_1 are compositions of primitive functions, $\mathcal{D}_{\psi,x}(g) := \mathcal{D}_{\psi_1,x} \circ \mathcal{D}_{\psi_2,\psi_1(x)}(g)$. By using this definition, we can characterize the gradient (computed by automatic differentiation) of a neural network defined in Eq. (1) as follows:

$$\begin{aligned} \mathcal{D}_{f,x}(g) &= \mathcal{D}_{\rho_1,x} \circ \mathcal{D}_{\sigma,y_1} \circ \dots \circ \mathcal{D}_{\sigma,y_{L-1}} \circ \mathcal{D}_{\rho_L,z_{L-1}}(g) \\ &= g \otimes A_L \otimes \sigma'(y_{L-1}) \otimes \dots \otimes \sigma'(y_1) \otimes A_1 \end{aligned}$$

where $y_l = \rho_l \circ \sigma \circ \dots \circ \sigma \circ \rho_1(x)$ and $z_l = \sigma \circ y_l$. We note that the operations in the second line *must be* computed in a left-associative way, unlike in the exact arithmetic case. Note that if $d_L = 1$, then $\mathcal{D}_{f,x}(1)$ corresponds to the

gradient of f . Furthermore, for a scalar-valued function ϕ_x and its derivative ϕ'_x , $\mathcal{D}_{f,x}(\phi'_x(f(x)))$ corresponds to $D^{\text{AD}}(\phi_x \circ f, x)$ introduced in Section 1.1. Throughout this paper, we refer to g as the *input gradient* (e.g., $\phi'_x(f(x))$ in Section 1.1) and $\mathcal{D}_{f,x}(g)$ as the *output gradient*.

Unlike classical derivatives under exact arithmetic, AD-computed gradients under floating-point arithmetic are not solely determined by the underlying function. Under exact arithmetic, two computational graphs representing the same differentiable function necessarily yield identical derivatives by the chain rule. In contrast, under floating-point arithmetic, intermediate rounding depends on the specific sequence of operations used in the computation. Consequently, two computational graphs representing the same function may produce different AD-computed gradients. For example, a linear mapping $x \mapsto ABx$ may be implemented either as two successive matrix multiplications or as a single multiplication by AB . Although these implementations define the same function, the gradients computed by AD can differ because intermediate rounding depends on the chosen computational graph.

3. Main Results

We are now ready to formally state our main results. To this end, we introduce the set of activation functions of interest

$$\Sigma = \{\lceil \text{ReLU} \rceil, \lceil \text{ELU} \rceil, \lceil \text{GELU} \rceil, \lceil \text{Swish} \rceil, \lceil \text{Sigmoid} \rceil, \lceil \tanh \rceil\}$$

and the following constants:

$$M_\sigma := \begin{cases} 2^{\epsilon_{\max}-2} & \text{if } \sigma \in \{\lceil \text{ReLU} \rceil, \lceil \text{ELU} \rceil, \lceil \text{GELU} \rceil, \lceil \text{Swish} \rceil\}, \\ 2^{\epsilon_{\max}-1} & \text{if } \sigma \in \{\lceil \text{Sigmoid} \rceil, \lceil \tanh \rceil\}, \end{cases}$$

$$H_\sigma := \begin{cases} \Omega & \text{if } \sigma \in \{\lceil \text{ReLU} \rceil, \lceil \text{ELU} \rceil, \lceil \text{GELU} \rceil, \lceil \text{Swish} \rceil\}, \\ 2^{M-1} & \text{if } \sigma = \lceil \text{Sigmoid} \rceil, \\ 2^{M-2} & \text{if } \sigma = \lceil \tanh \rceil. \end{cases}$$

Our main results are stated below. The proofs of Theorems 3.1 and 3.2 are in Appendices C.1 and C.2.

Theorem 3.1. *Let $E \geq 6$, $\sigma \in \Sigma$, $\mathcal{X} = [-M_\sigma, M_\sigma]_{\mathbb{F}}^d$, $f^* : \mathcal{X} \rightarrow \mathbb{F}$, $h^* : \mathcal{X} \rightarrow [-H_\sigma, H_\sigma]_{\mathbb{F}}$, and $g^* : \mathbb{F}^d \rightarrow \mathbb{F}^{1 \times d}$ such that $g^*(x) = 0$ for all $x \in \mathcal{X}$ with $h^*(x) = 0$. Then, for any $L \geq 7$, there exists an L -layer σ network f such that for all $x \in \mathcal{X}$,*

$$f(x) = f^*(x) \text{ and } \mathcal{D}_{f,x}(h^*(x)) = g^*(x).$$

Theorem 3.2. *Let $E \geq 6$, $\sigma \in \Sigma$, $\mathcal{X} = [-M_\sigma, M_\sigma]_{\mathbb{F}}^d$, $f^* : \mathcal{X} \rightarrow \mathbb{F}$, and $g^* : \mathbb{F}^d \times \mathbb{F} \rightarrow \mathbb{F}^{1 \times d}$ such that $g^*(x, -y) = -g^*(x, y)$. Then, for any $L \geq 2^{E+1} + 2M + 9$, there exists an L -layer σ network f such that for any $x \in \mathcal{X}$ and $y \in [-H_\sigma, H_\sigma]_{\mathbb{F}}$,*

$$f(x) = f^*(x) \text{ and } \mathcal{D}_{f,x}(y) = g^*(x, y).$$

Theorem 3.1 shows that 7-layer floating-point networks using $\sigma \in \Sigma$ can represent arbitrary function values and gradients on a wide domain $[-M_\sigma, M_\sigma]_{\mathbb{F}}^d$, when the input gradient $h^*(x)$ is bounded by H_σ (i.e., $\phi'_x(x)$ in Section 1.1); recall that if $h^*(x) = 1$, then $\mathcal{D}_{f,x}(h^*(x))$ corresponds to the gradient of f at x . This implies that floating-point networks can fit almost all function values and gradients. This result is extended to cover an arbitrary input gradient (i.e., $h^*(x)$ in Theorem 3.1) that may not be a function of the input x in Theorem 3.2, using more layers. Theorem 3.2 implies that there exists a single floating-point network f such that the gradient of $\phi_x \circ f$ computed by AD can output an arbitrary value, which is a function of $y = \phi'_x(f(x))$ for a large class of ϕ'_x .

In Theorems 3.1 and 3.2, we assume $E \geq 6$ in addition to Assumption 1. Nevertheless, our results cover floating-point formats using ≥ 16 bits, e.g., float32, float64, and bfloat16. We also present similar results for $4 \leq E \leq 5$ in Appendix C.3, which require more layers but cover 8bit formats (E5M2, E4M3).

In the remainder of this section, we sketch the main idea behind Theorem 3.1. We first describe our network construction for Theorem 3.1 in Section 3.1. Then, we introduce technical conditions on the activation functions and describe their implications in Section 3.2. Based on these conditions, we introduce technical lemmas and present generalizations of Theorems 3.1 and 3.2 to general activation functions in Section 3.3; Theorems 3.1 and 3.2 are corollaries of them.

3.1. Proof Sketch of Theorem 3.1

To prove Theorem 3.1, we construct two floating-point networks (say, f_1 and f_2) of the same depth. Here, f_1 represents the function values and has zero gradient, i.e., $f_1 = f^*$ and $\mathcal{D}_{f_1,x}(y) = 0$ for all x, y . f_2 does the exact opposite: $f_2(x) = 0$ and $\mathcal{D}_{f_2,x}(h^*(x)) = g^*(x)$ for all x . Then, $f = f_2 \# f_1$ satisfies the desired property in Theorem 3.1.

We construct f_1 as a composition of three networks: $f_1 = \psi_{13} \circ \psi_{12} \circ \psi_{11}$. Here, we design ψ_{11} so that

$$\psi_{11}(x) = (a \otimes \mathbb{1}_{x=z_1} \oplus b, \dots, a \otimes \mathbb{1}_{x=z_m} \oplus b), \quad (2)$$

where $z_1, \dots, z_m$ denote all inputs in the domain and a, b are some floats. Then, ψ_{12} suppresses the gradient so that

$$\mathcal{D}_{\psi_{13} \circ \psi_{12}, \psi_{11}(x)}(y) = (0, \dots, 0),$$

for all x, y by sequentially multiplying small weights. Here, by adding proper biases, ψ_{12} can preserve the output of ψ_{11} , i.e., $\psi_{12} \circ \psi_{11}$ is a vector of weighted indicator functions. Lastly, ψ_{13} performs a linear combination so that

$$f_1(x) = \psi_{13} \circ \psi_{12} \circ \psi_{11}(x) = \bigoplus_{i=1}^m f^*(x) \times \mathbb{1}_{x=z_i}.$$

We construct f_2 as a composition of two floating-point networks: $f_2 = \psi_{22} \circ \psi_{21}$. In the forward pass, $\psi_{21}(x) = (a^\dagger \otimes \mathbb{1}_{x=z_1} \oplus b^\dagger, \dots, a^\dagger \otimes \mathbb{1}_{x=z_m} \oplus b^\dagger)$ for some floats $a^\dagger$ and $b^\dagger$ as in ψ_{11} . Then, ψ_{22} suppresses this output so that $\psi_{22} \circ \psi_{21}(x) = 0$ for all x . The output of ψ_{21} is used in the backward pass. Here, ψ_{22} is designed to satisfy

$$\mathcal{D}_{\psi_{22}, \psi_{21}(x)}(y)_i = \begin{cases} 0 & \text{if } x \neq z_i, \\ y & \text{if } x = z_i. \end{cases} \quad (3)$$

Using this information, we further design the backward pass of ψ_{21} so that

$$\mathcal{D}_{\psi_{22} \circ \psi_{21}, x}(h^*(x)) = g^*(x).$$

We also use a similar strategy to prove Theorem 3.2, where the network corresponding to f_2 requires more complex operations.

3.2. Conditions on Activation Functions

In Section 3.1, we briefly sketched our construction of f in Theorem 3.1. There were three important parts in our construction. First, we construct indicator functions for all possible elements in the domain (ψ_{11} and ψ_{21}). We also have a network that preserves some specific gradients and suppresses others (ψ_{12} and ψ_{22}). Lastly, we transform an input non-zero gradient to the target gradient (ψ_{21}). In this section, we introduce conditions on general activation functions that suffice to implement these three components. To this end, we first introduce the following definition that is necessary to represent function values.

Definition 3.3 (Hwang et al. (2025b)). Let $\sigma : \mathbb{F} \rightarrow \mathbb{F}$, $d \in \mathbb{N}$, $\mathcal{X} \subset \mathbb{F}^d$, and $\mathcal{Y} \subset \mathbb{F}$. We say that “ $\mathcal{X}$ is σ -distinguishable with range $\mathcal{Y}$ ” if for every $x, x^\dagger \in \mathcal{X}$ with $x \neq x^\dagger$, there exists a floating-point affine transformation $\rho : \mathbb{F}^d \rightarrow \mathbb{F}$ such that

$$\sigma(\rho(x)) \neq \sigma(\rho(x^\dagger)) \quad \text{and} \quad \sigma(\rho(\mathcal{X})) \subset \mathcal{Y}.$$

It is known that to represent all functions from $\mathcal{X} \subset \mathbb{F}^d$ to $\mathbb{F}$ using σ networks, $\mathcal{X}$ should be distinguishable with range $\mathbb{F} \cup \{-\infty, \infty\}$ (see Lemma 3.2 in (Hwang et al., 2025b)). Hence, when we formally present our results, we will assume the distinguishability of the domain $\mathcal{X}$.

We now introduce technical conditions on general σ .

Condition 1. *There exist $\nu \in \mathbb{N}$, $\kappa \in \mathbb{N}_0$, $k_1, \dots, k_\nu \in \mathbb{Z}$, and $\gamma_0, \dots, \gamma_\nu \in [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{F}}$ with $|\gamma_i - \gamma_j| \leq 2^{\epsilon_{\max}}$ for all $i, j \in [\nu] \cup \{0\}$ such that*

- $\sigma(\gamma_0) = 0$, $|\sigma'(\gamma_i)|, |\sigma'(\gamma_0)| \leq 2^{-k_i} |\sigma(\gamma_i)| \forall i \in [\nu]$,
- $[\epsilon_{\min} - M, \epsilon_{\max} - M]_{\mathbb{Z}} \subset \bigcup_{i=1}^{\nu} [\epsilon_{\sigma(\gamma_i)} + \epsilon_{\min}, \epsilon_{\max} - \kappa + \min(\epsilon_{\sigma(\gamma_i)}, k_i)]_{\mathbb{Z}}$.

We use Condition 1 to represent the function values in f_1 while ensuring that the computed gradient does not overflow in the last layer. Specifically, we use $\sigma(\gamma_0) = 0$ to represent zero and $\sigma(\gamma_1), \dots, \sigma(\gamma_\nu)$ to represent non-zero function values in the last layer; the second bullet ensures that we can represent all values in $\mathbb{F}$. Here, if we use a large weight in the last layer, then we cannot accept a large input gradient $h^*(x)$ due to the overflow. κ in Condition 1 controls this issue: for a larger κ , we can use smaller weights in the last layer. Furthermore, to represent a target value v by $w \otimes \sigma(\gamma_i)$, $|w|$ should be proportional to $|v/\sigma(\gamma_i)|$. Hence, if $|\sigma'(\gamma_i)| \gg |\sigma(\gamma_i)|$, then overflow may occur for large $|h^*(x)|$ since the computed gradient in the last layer will be proportional to $|h^*(x)\sigma'(\gamma_i)v/\sigma(\gamma_i)|$. Thus, larger values of k_i and κ are preferable to avoid overflow during the evaluation of AD. Since ReLU and its variants can attain large output values with small gradients, they satisfy $\kappa \geq \epsilon_{\max} + 1$. On the other hand, Sigmoid and tanh have $\kappa \approx M$ due to their bounded function values.

Our next condition is used to suppress undesired gradients.

Condition 2. *There exist $\delta_0, \delta_1 \in \mathbb{F}$ with $|\delta_0|, |\delta_1|, |\delta_0 - \delta_1| \leq 2^{\epsilon_{\max}}$, and $\eta \in \mathbb{N}$ such that*

- $|\sigma(\delta_0)|, |\sigma(\delta_1)|, |\sigma(\delta_0) - \sigma(\delta_1)| \leq 2^{\epsilon_{\max}}, \sigma(\delta_0) \neq \sigma(\delta_1)$,
- $-2^{-M-1} \times \sigma'(\delta_1) \leq 2\sigma'(\delta_0) \leq \sigma'(\delta_1)$,
- $2^{-M} \leq |\sigma'(\delta_1)| = 2^{\epsilon_{\sigma'(\delta_1)}} \leq 1$,
- $-\epsilon_{\sigma'(\delta_1)} < \eta \leq \epsilon_{\max}, |\sigma(\delta_0)|, |\sigma(\delta_1)| \leq 2^{\epsilon_{\max}-\eta}$.

We can multiply appropriate weights to control the magnitude of gradients without overflow: choosing $w = 2^e$ for $e \in [\epsilon_{\min}, \eta]_{\mathbb{Z}}$ guarantees $w \otimes \sigma(\delta_0), w \otimes \sigma(\delta_1) \in \mathbb{F}$. By repeatedly applying this strategy, multiplying appropriate weights, and adding appropriate biases, we can implement a network that increases or decreases the computed gradients and controls them depending on the input, as in Eq. (3). For $\sigma \in \Sigma$, there exist δ_0, δ_1 such that $\sigma(\delta_0) \approx 0$ and $|\sigma(\delta_1)| \in [1/4, 4]$. Hence, for such σ , we have $\eta \approx \epsilon_{\max}$ since $\max(|\sigma(\delta_1)|, |\sigma(\delta_0)|) \in [1/4, 4]$.

Condition 3. *There exists $\zeta \in (0, (2^{\epsilon_{\max}})^-]_{\mathbb{F}}$ such that at least one of the following conditions holds:*

- *There exists $x \in \mathbb{F} \setminus (-\zeta, \zeta)_{\mathbb{F}}$ such that $\zeta \oplus |x| \in \mathbb{F}$, $|\sigma'(x)| \geq 2^{-M}$, x is even,¹ and*

$$\sigma([-(\zeta \oplus |x|), (\zeta \oplus |x|)]_{\mathbb{F}}) \subset [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}].$$
- $|\sigma'(0)| \geq 2^{-M}$ and $\sigma([-2\zeta, 2\zeta]_{\mathbb{F}}) \subset [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$.

Condition 3 is a technical condition used to create the desired gradient value in the first layer of f_2 . In particular, if the condition in the first bullet is satisfied, we map each

¹See Section 2.2 for the definition of “even”.

coordinate of the input in $[-\zeta, \zeta]_{\mathbb{F}}$ to x so that the resulting gradient in the backward pass is not too small (it will be multiplied by $|\sigma'(x)| \geq 2^{-M}$). Here, we additionally use $\zeta \oplus |x| \in \mathbb{F}$ to avoid overflow in the first layer while keeping the activation values bounded. If the condition in the second bullet is satisfied, we can similarly map each coordinate of the input to zero.

3.3. Results for General Activation Functions

In this section, we introduce lemmas that are used to construct f_1 and f_2 , which were introduced in Section 3.1 using Conditions 1–3. Then, using these lemmas, we derive general versions of Theorems 3.1 and 3.2 that hold for general activation functions. To better describe our results, given an activation function σ satisfying Conditions 1–3, we define

$$\tau := \left\lceil \max \left(\frac{2^E + M}{\eta + \epsilon_{\sigma'(\delta_1)}} + 4, \frac{2^E + M}{-\epsilon_{\min} - \epsilon_{\sigma'(\delta_1)}} + 4, 7 \right) \right\rceil_{\mathbb{Z}}, \quad (4)$$

where δ_1 and η are from Condition 2.

The following lemmas are for f_1 and f_2 . We present the proofs of Lemmas 3.4 and 3.5 in Sections 4.1 and 4.2.

Lemma 3.4. *Suppose σ satisfies Conditions 1–2 and $\mathcal{X} \subset [-\zeta, \zeta]_{\mathbb{F}}^d$ is σ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. Let $f^* : \mathcal{X} \rightarrow \mathbb{F}$. Then, for any $L \geq 5$, there exists an L -layer σ network f such that for any $x \in \mathbb{F}^d$ and $y \in [-2^{\kappa}, 2^{\kappa}]_{\mathbb{F}}$, and L -layer σ network $\tilde{f} : \mathcal{X} \rightarrow \mathbb{F}$,*

$$f(x) = f^*(x), \mathcal{D}_{f,x}(y) = \mathbf{0}_d^{\top}, \text{ and } \mathcal{D}_{\tilde{f}\#f,x}(y) = \mathcal{D}_{\tilde{f},x}(y).$$

Lemma 3.5. *Suppose σ satisfies Conditions 1–3 and $\mathcal{X} \subset [-\zeta, \zeta]_{\mathbb{F}}^d$ is σ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. Let $g^* : \mathcal{X} \rightarrow \mathbb{F}^{1 \times d}$ and $h^* : \mathbb{F}^d \rightarrow [-\Omega \cdot 2^{\epsilon_{\sigma'(\delta_1)}}, \Omega \cdot 2^{\epsilon_{\sigma'(\delta_1)}}]_{\mathbb{F}}$ such that $g^*(x) = 0$ for all $x \in \mathcal{X}$ with $h^*(x) = 0$. Then, for any $L \geq \tau$, there exists an L -layer σ network f such that, for any $x \in \mathcal{X}$ and L -layer σ network $\tilde{f} : \mathcal{X} \rightarrow \mathbb{F}$,*

$$f(x) = 0, \mathcal{D}_{f,x}(h^*(x)) = g^*(x), \text{ and } (f\#\tilde{f})(x) = \tilde{f}(x).$$

The network in Lemma 3.4 is for f_1 , where three layers are used to construct the indicator function (ψ_{11}), and the remaining layers are used to suppress gradients ($\psi_{13} \circ \psi_{12}$). The network in Lemma 3.5 is for f_2 . In this case, we use two layers to control gradients as in Eq. (3) (ψ_{22}) and use the rest to construct the target gradient (ψ_{21}).

Let f_1 and f_2 be σ networks of $\geq \tau$ layers in Lemmas 3.4 and 3.5, respectively. Then, by choosing $f = f_2\#f_1$, Theorem 3.6 follows, which is a generalization of Theorem 3.1.

Theorem 3.6. *Suppose σ satisfies Conditions 1–3 and $\mathcal{X} \subset [-\zeta, \zeta]_{\mathbb{F}}^d$ is σ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. Let $f^*, h^* : \mathcal{X} \rightarrow \mathbb{F}$, $g^* : \mathcal{X} \rightarrow \mathbb{F}^{1 \times d}$ such that $|h^*(x)| \leq \min(\Omega \times 2^{\sigma'(\delta_1)}, 2^{\kappa})$ for all $x \in \mathcal{X}$ and $g^*(x) = 0$ for all*

$x \in \mathcal{X}$ with $h^(x) = 0$. Then, for any $L \geq \tau$, there exists an L -layer σ network f such that for any $x \in \mathcal{X}$,*

$$f(x) = f^*(x) \text{ and } \mathcal{D}_{f,x}(h^*(x)) = g^*(x).$$

To prove Theorem 3.2, we construct a variant of f_2 that accepts an arbitrary input gradient using additional layers, as illustrated in the following lemma. The proof of Lemma 3.7 is presented in Appendix B.4.

Lemma 3.7. *Suppose σ satisfies Conditions 1–3 and $\mathcal{X} \subset [-\zeta, \zeta]_{\mathbb{F}}^d$ is σ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. Let $g^* : \mathcal{X} \times \mathbb{F} \rightarrow \mathbb{F}^{1 \times d}$, such that $g^*(x, -y) = -g^*(x, y)$. Then, for any $L \geq 2^{E+1} + 2M + 2 + \tau$, there exists an L -layer σ network f such that for any $x \in \mathcal{X}$, $y \in [-\Omega \cdot 2^{\epsilon_{\sigma'(\delta_1)}}, \Omega \cdot 2^{\epsilon_{\sigma'(\delta_1)}}]_{\mathbb{F}}$, and L -layer σ network $\tilde{f} : \mathcal{X} \rightarrow \mathbb{F}$,*

$$f(x) = 0, \mathcal{D}_{f,x}(y) = g^*(x, y), \text{ and } (f\#\tilde{f})(x) = \tilde{f}(x).$$

By combining Lemmas 3.4 and 3.7, we can prove the following theorem.

Theorem 3.8. *Suppose σ satisfies Conditions 1–3 and $\mathcal{X} \subset [-\zeta, \zeta]_{\mathbb{F}}^d$ is σ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. Let $f^* : \mathcal{X} \rightarrow \mathbb{F}$ and $g^* : \mathcal{X} \times \mathbb{F} \rightarrow \mathbb{F}^{1 \times d}$ such that $g^*(x, -y) = -g^*(x, y)$. Then, for any $L \geq 2^{E+1} + 2M + 2 + \tau$, there exists an L -layer σ network f such that for any $x \in \mathcal{X}$ and $y \in \mathbb{F}$ with $|y| \leq \min(\Omega \cdot 2^{\epsilon_{\sigma'(\delta_1)}}, 2^{\kappa})$,*

$$f(x) = f^*(x) \text{ and } \mathcal{D}_{f,x}(y) = g^*(x, y).$$

We note that Theorems 3.1 and 3.2 can be derived from Theorems 3.6 and 3.8 by using constants in Table 1.

Table 1. Constants in Conditions 1–3. ReLU* denotes ReLU, ELU, GELU, and Swish. See Lemmas C.2–C.4 for derivations.

$\hat{\sigma}(x)$	κ	$\sigma'(\delta_1)$	η	ζ
ReLU*	$\epsilon_{\max} - 1$	1	$2^{E-1} - 2$	$2^{\epsilon_{\max}-2}$
Sigmoid	$M - 1$	1/4	$2^{E-1} - 2$	$2^{\epsilon_{\max}-1}$
tanh	$M - 2$	1	$2^{E-1} - 1$	$2^{\epsilon_{\max}-1}$

We also implement our constructions in Lemma 3.4 and Lemma 3.5 for ReLU and observe its correctness.²

Although we assumed that the gradient through an activation function follows the rule $g \mapsto g \otimes \sigma'(x)$ (see Section 2.4), the discussion can be extended to a more general setup in which the gradient is computed as an arbitrary elementwise function. For example, if σ is the sigmoid function, then a typical implementation is $g \mapsto g \otimes \sigma'(x) \otimes (1 \ominus \sigma'(x))$. In this case, by slightly modifying conditions, we can obtain analogous results. For example, in Condition 1, which

²The code is available at <https://github.com/yechnan/fp-grad-rep>.

will be presented in the next section, $\sigma'(\gamma_i) \leq 2^{-k}|\sigma(\gamma_i)|$ should be replaced with $|\mathcal{D}_{\sigma, \gamma_i}(g)| \leq g \otimes 2^{-k}|\sigma(\gamma_i)|$. Other conditions can be refined accordingly.

4. Proofs

4.1. Proof of Lemma 3.4

In this section, we prove Lemma 3.4 by explicitly constructing the target σ network f (i.e., f_1 in Section 3.1). Let $\{z_1, \dots, z_m\} = \mathcal{X}$. We will construct f as follows:

$$f(x) = \#_{i=1}^m \lambda_i(x), \quad \lambda_i : (z_i, \setminus z_i) \mapsto (f^*(z_i), 0),$$

where $\lambda_i : \mathcal{X} \rightarrow \overline{\mathbb{F}}$ are σ networks with the same number of layers. Here, we will properly design the last layer of λ_i so that $\#_{i=1}^m \lambda_i(x) = f^*(x)$ in the target domain $\mathcal{X}$.

As illustrated in Section 3.1, our construction of λ_i consists of three parts: $\lambda_i = \psi_{i,13} \circ \psi_{i,12} \circ \psi_{i,11}$. In particular, we use the following lemma to construct $\psi_{i,11}$. The proof of Lemma 4.1 is presented in Appendix B.2.

Lemma 4.1. *Let $d \in \mathbb{N}$. Suppose that σ satisfies Condition 1 and that $\mathcal{X} \subset \mathbb{F}^d$ is σ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. Then, for any $z \in \mathbb{F}^d$ and any $c \in \{\delta_0, \delta_1\}$, there exists a three-layer σ network $f : \mathcal{X} \rightarrow \mathbb{F}^2$ ending with the activation function such that*

- $f(x) = [\sigma(c) \mathbb{1}_{z=x} \quad \sigma(c) \mathbb{1}_{z=x}]$ and
- $\mathcal{D}_{f,x}([g \quad -g]) = 0$ for all $x \in \mathcal{X}$ and $g \in \mathbb{F}$.

By Lemma 4.1, there exists a three-layer σ network $\psi_{i,11}$ ending with the activation function such that

$$\psi_{i,11} : (z_i, \setminus z_i) \mapsto ([\sigma(\delta_1) \quad \sigma(\delta_1)], [\sigma(\delta_0) \quad \sigma(\delta_0)]),$$

$$\text{and } \mathcal{D}_{\psi_{i,11},x}([g \quad -g]) = 0.$$

We next construct $\psi_{i,13}$ and $\psi_{i,12}$ using the following lemmas. The proofs of Lemmas 4.2 and 4.3 are presented in Appendices B.3 and B.5.

Lemma 4.2. *Suppose σ satisfies Conditions 1 and 2. For any $y_1, y_2 \in \mathbb{F}$ such that $|y_1|, |y_2|, |y_1 - y_2| \leq 2^{\epsilon_{\max}}$, there exists a one-layer σ network $f : \mathbb{F}^2 \rightarrow \mathbb{F}$ ending with the activation function such that*

- $f : ([\sigma(\delta_0) \quad \sigma(\delta_0)], [\sigma(\delta_1) \quad \sigma(\delta_1)]) \mapsto (\sigma(y_1), \sigma(y_2))$,
- $\mathcal{D}_{f,x}(g) = [g_1 \quad -g_1]$ for $x = \sigma(\delta_1), \sigma(\delta_0)$ and for all g satisfying $g \otimes \sigma'(y_1), g \otimes \sigma'(y_2) \in \mathbb{F}$ for some $g_1 \in \mathbb{F}$ depending on g .

Lemma 4.3. *Suppose σ satisfies Condition 1. Then, for any $c \in \mathbb{F}$, there exist $n \in \mathbb{N}$, $c_i \in \{\gamma_0, \dots, \gamma_\nu\}$, and $w_i \in \mathbb{F}$ for $i \in [n]$ such that for any $g \in [-2^\kappa, 2^\kappa]_{\mathbb{F}}$,*

$$c = \bigoplus_{i=1}^n w_i \otimes \sigma(c_i)$$

$$\text{and } g \otimes w_i \otimes \sigma'(c_i), g \otimes w_i \otimes \sigma'(\gamma_0) \in \mathbb{F}.$$

By Lemma 4.3, there exist $n_i \in \mathbb{N}$, $w_{i,1}, \dots, w_{i,n_i} \in \mathbb{F}$, and $c_{i,1}, \dots, c_{i,n_i} \in \{\gamma_1, \dots, \gamma_\nu\}$ such that

$$f^*(z_i) = \bigoplus_{j=1}^{n_i} (w_{i,j} \otimes \sigma(c_{i,j})).$$

By Lemma 4.2, there exists a two layer network $\psi_{i,12}^j$ satisfying the followings:

$$\psi_{i,12}^j : ([\sigma(\delta_0) \quad \sigma(\delta_0)], [\sigma(\delta_1) \quad \sigma(\delta_1)]) \mapsto (\sigma(\gamma_0), \sigma(c_{i,j}))$$

and

$$\mathcal{D}_{\psi_{i,12}^j \circ \psi_{i,11},x}(g) = 0$$

for $x = \sigma(\delta_1), \sigma(\delta_0)$ and for all g satisfying $g \otimes \sigma'(y_1), g \otimes \sigma'(y_2) \in \mathbb{F}$. By defining $\psi_{i,12} : \mathbb{F} \rightarrow \mathbb{F}^{n_i}$ as $\psi_{i,12} = (\psi_{i,12}^1, \dots, \psi_{i,12}^{n_i})$, we get a σ network $\psi_{i,12} : \mathbb{F} \rightarrow \mathbb{F}^{n_i}$ ending with the activation function such that

$$\begin{aligned} \psi_{i,12}(\delta_0) &= (\sigma(\gamma_0), \dots, \sigma(\gamma_0)), \\ \psi_{i,12}(\delta_1) &= (\sigma(c_{i,1}), \dots, \sigma(c_{i,n_i})). \end{aligned} \quad (5)$$

and $\mathcal{D}_{\psi_{i,12} \circ \psi_{i,11}, \sigma(\delta_0)}(g) = \mathcal{D}_{\psi_{i,12} \circ \psi_{i,11}, \sigma(\delta_1)}(g) = 0$ for all $g \in \mathbb{F}^{1 \times n_i}$ satisfying $g_k \otimes \sigma'(\gamma_0), g_k \otimes \sigma'(c_{i,k}) \in \mathbb{F}$ for $k \in [n_i]$. We lastly define $\psi_{i,13} : \mathbb{F}^{n_i} \rightarrow \overline{\mathbb{F}}$ as an affine transformation:

$$\psi_{i,13}(x) = (w_{i,1}, \dots, w_{i,n_i})^\top \otimes x. \quad (6)$$

As we introduced at the beginning of the proof, we construct λ_i as $\lambda_i = \psi_{i,13} \circ \psi_{i,12} \circ \psi_{i,11}$. Then, by the definitions of $\psi_{i,11}$, $\psi_{i,12}$, and $\psi_{i,13}$ (Section 4.1 and Eqs. (5) and (6)), we have $\lambda_i = f^*(z_i) \times \mathbb{1}_{x=z_i}$ on $\mathcal{X}$. Furthermore, one can observe that

$$\mathcal{D}_{\psi_{i,13} \circ \psi_{i,12}, \delta_0}(g) = \mathcal{D}_{\psi_{i,13} \circ \psi_{i,12}, \delta_1}(g) = 0$$

for all $g \in [-2^\kappa, 2^\kappa]_{\mathbb{F}}$, i.e., $\mathcal{D}_{\lambda_i, x}(g) = 0$ for all $x \in \mathcal{X}$ and $g \in [-2^\kappa, 2^\kappa]_{\mathbb{F}}$. We now construct the target network as $f = \#_{i=1}^m \lambda_i$. Then, from our construction of λ_i , we have

$$f(x) = f^*(x), \quad \mathcal{D}_{f,x}(g) = 0, \quad \text{and } \mathcal{D}_{f \# \tilde{f}, x}(g) = \mathcal{D}_{\tilde{f}, x}(g)$$

for all $x \in \mathcal{X}$, $g \in [-2^\kappa, 2^\kappa]_{\mathbb{F}}$, and σ networks $\tilde{f} : \mathcal{X} \rightarrow \overline{\mathbb{F}}$ of the same number of layers. Since $\psi_{i,11}$ uses three layers, $\psi_{i,12}$ uses one layer, and $\psi_{i,13}$ uses one layer, for any $L \geq 5$, f can be an L -layer σ network. This completes the proof of Lemma 3.4.

4.2. Proof of Lemma 3.5

Let $\{z_1, \dots, z_m\} = \mathcal{X}$. As in the proof of Lemma 3.4, we construct the target network f (i.e., f_2 in Section 3.1) as $f = \#_{i=1}^m \lambda_i$, where each λ_i satisfies that for any $x \in \mathcal{X}$,

$$\lambda_i(x) = 0 \quad \text{and} \quad \mathcal{D}_{\lambda_i, x}(h^*(x)) = g^*(z_i) \times \mathbb{1}_{x=z_i}. \quad (7)$$

Following our proof sketch in Section 3.1, our construction of λ_i consists of two parts: $\lambda_i = \psi_{i,22} \circ \psi_{i,21}$, where $\psi_{i,22}$ and $\psi_{i,21}$ are designed so that for $j \in \{0, 1\}$,

- (1) $\psi_{i,21} : (z_i, \setminus z_i) \mapsto (\delta_1, \delta_0)$, $\mathcal{D}_{\psi_{i,21},x}(h_1^*(x)) = g^*(x)$,
- (2) $\psi_{i,22}(\delta_j) = 0$, $\mathcal{D}_{\psi_{i,22},\psi_{i,21}(x)}(h^*(x)) = h_1^*(x) \times \mathbb{1}_{x=z_i}$,

where $h_1 : \mathbb{F}^d \rightarrow \mathbb{F}$ is a function satisfying $0 < |h_1(x)| \leq |h(x)|$ whenever $h(x) \neq 0$, and $h_1(x) = 0$ if $h(x) = 0$. In particular, we construct $\psi_{i,21}$ as $\psi_{i,21} = \pi_{i,2} \# \pi_{i,1}$ where $\pi_{i,1}$ and $\pi_{i,2}$ are σ networks satisfying

- (1-1) $\pi_{i,1} : (z_i, \setminus z_i) \mapsto (\delta_1, \delta_0)$, $\mathcal{D}_{\pi_{i,1},x}(y) = 0$ for all $x \in \mathcal{X}$ and $y \in [-2^\kappa, 2^\kappa]_{\mathbb{F}}$,
- (1-2) $\pi_{i,2}(x) = 0$, $\mathcal{D}_{\pi_{i,2},x}(h_1^*(x)) = g^*(x)$ for all $x \in \mathcal{X}$.

By Lemma 3.4, there exists such a σ network $\pi_{i,1}$.

We construct $\pi_{i,2}$ using the following lemmas. The proofs of Lemmas 4.4 and 4.5 are in Appendices B.6 and B.7.

Lemma 4.4. *Suppose σ satisfies Condition 2. Then, for any $n \in \mathbb{N}$, $y^* \in \mathbb{F}^{1 \times n}$, $y \in \mathbb{F}$ with $0 < |y| \leq \Omega \cdot 2^{\epsilon_{\sigma'}(\delta_1)}$ and $L \geq \left\lceil \max \left(\frac{2^E + M}{\eta + \epsilon_{\sigma'}(\delta_1)}, \frac{2^E + M}{-\epsilon_{\min} - \epsilon_{\sigma'}(\delta_1)} \right) \right\rceil_{\mathbb{Z}}$, there exists an L -layer σ network $f : \mathbb{F}^n \rightarrow \mathbb{F}$ starting with σ such that*

$$f : \delta_1 \times \mathbf{1}_n \mapsto 0 \text{ and } \mathcal{D}_{f,\delta_1 \times \mathbf{1}_n}(y) = y^*.$$

Lemma 4.5. *Suppose σ satisfies Condition 3 and $\mathcal{X} \subset [-\zeta, \zeta]_{\mathbb{F}}^d$ is σ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. Then, for any $y^* \in \mathbb{F}^{1 \times d}$ and $z \in \mathcal{X}$, there exist $n \in \mathbb{N}$, $y \in [-\Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}, \Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}]_{\mathbb{F}}^{1 \times n}$, and a two-layer σ network $f : \mathbb{F}^d \rightarrow \mathbb{F}^n$ such that for any $x \in [-\zeta, \zeta]_{\mathbb{F}}^d$,*

$$f(x) = \delta_1 \times \mathbf{1}_n \text{ and } \mathcal{D}_{f,z}(y) = y^*.$$

By Lemmas 4.4 and 4.5, there exists a σ network $\pi_{i,2}$ that satisfies (1-2). Then, since $\pi_{i,1}$ is constructed using Lemma 3.4, $\psi_{i,21} = \pi_{i,2} \# \pi_{i,1}$ satisfies (1) for all $x \in \mathcal{X}$.

Lastly, we construct $\psi_{i,22}$ using the following lemma. The proof of Lemma 4.6 is presented in Appendix B.8.

Lemma 4.6. *Suppose σ satisfies Conditions 1 and 2. Let $c_1, c_2 \in \mathbb{F}$ be such that $|c_1|, |c_2|, |c_1 - c_2| \leq 2^{\epsilon_{\max}}$. Then, there exists a two-layer σ network $f : \mathbb{F} \rightarrow \mathbb{F}$ starting with σ such that for any $y \in [-\Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}, \Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}]_{\mathbb{F}}$,*

$$f : (\delta_0, \delta_1) \mapsto (c_1, c_2), \mathcal{D}_{f,\delta_0}(y) = 0, \text{ and } \mathcal{D}_{f,\delta_1}(y) = y_1,$$

where $y_1 \in \mathbb{F}$ is a constant satisfying $0 < |y_1| \leq |y|$ whenever $y \neq 0$, and $y_1 = 0$ if $y = 0$.

By Lemma 4.6, there exists a σ network $\psi_{i,22}$ starting with the activation function such that for any $j \in \{0, 1\}$ and $y \in [-\Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}, \Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}]_{\mathbb{F}}$,

$$\psi_{i,22}(\delta_j) = 0 \text{ and } \mathcal{D}_{\psi_{i,22},\delta_j}(y) = y_1 \times \mathbb{1}_{j=1}.$$

Hence, together with our $\psi_{i,21}$, (2) is satisfied for all $x \in \mathcal{X}$, i.e., $\lambda_i = \psi_{i,22} \circ \psi_{i,21}$ satisfies Eq. (7) on $\mathcal{X}$. Furthermore, from our construction of λ_i , $f = \#_{i=1}^m \lambda_i$ satisfies

$$f(x) = 0, \mathcal{D}_{f,x}(h_1^*(x)) = g^*(x), \text{ and } (f \# \tilde{f})(x) = \tilde{f}(x)$$

for all $x \in \mathcal{X}$ and σ networks $\tilde{f}$ with the same depth. Specifically, we note that $\mathcal{D}_{f,x}(h_1^*(x)) = g^*(x)$ follows from our construction of $\psi_{i,22}$ where $\mathcal{D}_{\psi_{i,22},\psi_{i,21}(x)} = 0$ if $x \neq z_i$; hence, $\mathcal{D}_{f,x}(h_1^*(x))$ can be represented by the sum of zeros and $\mathcal{D}_{\lambda_k,x}(h_1^*(x)) = g^*(x)$ for $x = z_k$. We also note that f can be constructed by using $\geq \tau$ layers; see our constructions of $\pi_{i,1}$, $\pi_{i,2}$, $\psi_{i,22}$, and Lemmas 3.4 and 4.4–4.6. This completes the proof of Lemma 3.5.

5. Conclusion

In this work, we show that floating-point neural networks can represent arbitrary function values and gradients computed via automatic differentiation. We further extend this result to the case where the target gradient is a function of ϕ'_x , and ϕ_x denotes the function composed with the network. Our results show the discrepancy between exact and floating-point arithmetic: the target gradient should be proportional to the input gradient under exact arithmetic, but this is not the case under floating-point arithmetic. We believe our results help in understanding the properties of real implementations of neural networks.

Acknowledgement

We thank Reviewer NB9Q for the insightful suggestion that helped reduce the number of layers in our construction. This work was supported by the National Research Foundation of Korea (NRF) grant (No. RS-2024-00406127, RS-2024-00415812, RS-2025-00515264, RS-2025-24534003, RS-2026-25472534), Institute of Information & communications Technology Planning & Evaluation (IITP) grant (No. RS-2019-II190079, 2019-0-01842) funded by the Korea government (MSIT), and Global University Project grant funded by the GIST in 2025.

Impact Statement

This paper analyzes the representation power of floating-point neural networks. We could not find any notable potential societal consequences of our work.

References

- Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G. S., Davis, A., Dean, J., Devin, M., et al. Tensorflow: Large-scale machine learning on heterogeneous distributed systems. *arXiv preprint arXiv:1603.04467*, 2016.

- Bora, A., Jalal, A., Price, E., and Dimakis, A. G. Compressed sensing using generative models. In *International Conference on Machine Learning (ICML)*, 2017.
- Czarnecki, W. M., Osindero, S., Jaderberg, M., Swirszcz, G., and Pascanu, R. Sobolev training for neural networks. In *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2017.
- Fredrikson, M., Jha, S., and Ristenpart, T. Model inversion attacks that exploit confidence information and basic countermeasures. In *Proceedings of the 22nd ACM SIGSAC conference on computer and communications security*, pp. 1322–1333, 2015.
- Gatys, L. A., Ecker, A. S., and Bethge, M. Image style transfer using convolutional neural networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- Goodfellow, I. J., Shlens, J., and Szegedy, C. Explaining and harnessing adversarial examples. In *International Conference on Learning Representations (ICLR)*, 2015.
- Google. Improve your model’s performance with bfloat16. <https://cloud.google.com/tpu/docs/bfloat16>.
- Griewank, A. and Walther, A. *Evaluating derivatives: principles and techniques of algorithmic differentiation*. SIAM, 2008.
- Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., and Courville, A. C. Improved Training of Wasserstein GANs. In *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2017.
- Hwang, G. Optimal minimum width for the universal approximation of continuously differentiable functions by deep narrow mlps. In *Annual Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- Hwang, G., Lee, W., Park, Y., Park, S., and Saad, F. Floating-point neural networks are provably robust universal approximators. In *International Conference on Computer Aided Verification*, pp. 301–326. Springer, 2025a.
- Hwang, G., Park, Y., Lee, W., and Park, S. Floating-point neural networks can represent almost all floating-point functions. In *Forty-second International Conference on Machine Learning*, 2025b.
- IEEE. *IEEE Standard for Floating-Point Arithmetic (IEEE Std 754-2019)*. IEEE, Piscataway, NJ, USA, 2019. doi: 10.1109/IEEESTD.2019.8766229.
- Li, X. Simultaneous approximations of multivariate functions and their derivatives by neural networks with one hidden layer. *Neurocomputing*, 12(4):327–343, 1996.
- Lillicrap, T. P., Hunt, J. J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D., and Wierstra, D. Continuous control with deep reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2016.
- Lin, H. and Jegelka, S. Resnet with one-neuron hidden layers is a universal approximator. 2018.
- Micikevicius, P., Stolic, D., Burgess, N., Cornea, M., Dubey, P., Grisenthwaite, R., Ha, S., Heinecke, A., Judd, P., Kamalu, J., et al. FP8 formats for deep learning. *arXiv preprint arXiv:2209.05433*, 2022.
- Park, S., Park, Y., and Hwang, G. On the expressive power of floating-point transformers. *arXiv preprint arXiv:2601.16450*, 2026.
- Park, Y., Hwang, G., Lee, W., and Park, S. Expressive power of ReLU and step networks under floating-point operations. *Neural Networks*, 175:106297, 2024.
- Pinkus, A. Approximation theory of the MLP model in neural networks. *Acta Numerica*, 8:143 – 195, 1999.
- Raissi, M., Perdikaris, P., and Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational physics*, 378:686–707, 2019.
- Schäfer, A. M. and Zimmermann, H.-G. Recurrent neural networks are universal approximators. *International Journal of Neural Systems*, 17(04):253–263, 2007.
- Simonyan, K., Vedaldi, A., and Zisserman, A. Deep inside convolutional networks: Visualising image classification models and saliency maps. In *Workshop at International Conference on Learning Representations*, 2014.
- Yun, C., Bhojanapalli, S., Rawat, A. S., Reddi, S., and Kumar, S. Are transformers universal approximators of sequence-to-sequence functions? In *International Conference on Learning Representations (ICLR)*, 2020.
- Zhou, D.-X. Universality of deep convolutional neural networks. *Applied and Computational Harmonic Analysis*, 48(2):787–794, 2020.

A. Additional Preliminaries

A.1. Additional Notations

If $x \in \mathbb{F}$ and $|x| \geq 2^{\epsilon_{\min}}$, we define $\mathfrak{s}_x \in \{-1, 1\}$, $\mathfrak{m}_x \in \{1 + i/2^M \mid i \in [0, 2^M - 1]_{\mathbb{Z}}\}$, and $\mathfrak{e}_x \in [\epsilon_{\min}, \epsilon_{\max}]_{\mathbb{Z}}$ be the unique numbers satisfying $x = \mathfrak{s}_x \times \mathfrak{m}_x \times 2^{\mathfrak{e}_x}$. For $|x| < 2^{\epsilon_{\min}}$, we define $\mathfrak{s}_x \in \{-1, 1\}$ and $\mathfrak{m}_x \in \{i/2^M \mid i \in [0, 2^M - 1]_{\mathbb{Z}}\}$ be the unique numbers satisfying $x = \mathfrak{s}_x \times \mathfrak{m}_x \times 2^{\epsilon_{\min}}$; in this case, $\mathfrak{e}_x = \epsilon_{\min}$. We also define $\mathfrak{m}_{x,0}, \dots, \mathfrak{m}_{x,M} \in \{0, 1\}$ as the unique values satisfying $\mathfrak{m}_x = \sum_{i=0}^M \mathfrak{m}_{x,i} 2^{-i}$.

A.2. Floating-Point Operations on Non-Finite Floats

For $x = \text{NaN}$ and $y \in \overline{\mathbb{F}}$, we define $x \oplus y = y \oplus x = x \otimes y = y \otimes x = \text{NaN}$. For $x = \pm\infty$ and $y \in \mathbb{F}$, we define $x \oplus y = y \oplus x = \pm\infty$. We also define $\infty \oplus (-\infty) = (-\infty) \oplus \infty = \text{NaN}$, $\infty \oplus \infty = \infty$, $(-\infty) \oplus (-\infty) = -\infty$. We define $x \ominus y = x \oplus (-y)$ where $-\text{NaN} = \text{NaN}$ and $-(\pm\infty) = \mp\infty$. For $x = \infty$, $0 < y \in \mathbb{F} \cup \{\infty\}$, and $\mathfrak{s}_x, \mathfrak{s}_y \in \{-1, 1\}$, we define $(\mathfrak{s}_{xx}) \otimes (\mathfrak{s}_{yy}) = (\mathfrak{s}_{yy}) \otimes (\mathfrak{s}_{xx}) = (\mathfrak{s}_x \mathfrak{s}_y) \infty$. For $x = \pm\infty$ and $y = 0$, we define $x \otimes y = y \otimes x = \text{NaN}$.

For $\hat{\sigma} : \mathbb{R} \rightarrow \mathbb{R}$, we define $\lceil \hat{\sigma} \rceil(\text{NaN}) = \text{NaN}$ and the values of $\lceil \hat{\sigma} \rceil$ at infinity is the rounded value of the right limit of $\hat{\sigma}$ if it exists; otherwise, it is NaN. $\lceil \hat{\sigma} \rceil(-\infty)$ is also defined in a similar way.

A.3. Additional Definitions and Technical Lemmas

We adopt the definitions of *sequential addition* and *transferability* from Hwang et al. (2025b), with a slight modification.

Definition A.1 (Sequential addition (Hwang et al., 2025b)). Let $\sigma : \overline{\mathbb{F}} \rightarrow \overline{\mathbb{F}}$ and $\mathcal{S}_\sigma := \{w \otimes \sigma(c) \mid w, c \in \mathbb{F} \text{ with } w \otimes \sigma(c) \in \mathbb{F}\}$. We say that a function $f : \overline{\mathbb{F}} \rightarrow \overline{\mathbb{F}}$ is a “sequential addition using σ ” if there exist $n \in \mathbb{N}$ and $z_1, \dots, z_n \in \mathcal{S}_\sigma$ such that for each $x \in \mathbb{F}$,

$$f(x) = x \oplus z_1 \oplus \dots \oplus z_n.$$

We often drop σ and use $\mathcal{S}$ to denote $\mathcal{S}_\sigma$ when it is clear from the context.

Unlike Hwang et al. (2025b), we omit the condition $f(\mathbb{F}) \subset \mathbb{F}$. Sequential additions are closed under composition with neural networks. In particular, if s is a sequential addition using σ and $f : \mathbb{F}^d \rightarrow \overline{\mathbb{F}}$ is a σ -neural network, then $s \circ f$ is again a σ -neural network. Note that composing with a sequential addition does not alter the output gradient.

Definition A.2 (Transferability (Hwang et al., 2025b)). Let $n \in \mathbb{N}$ and $(x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbb{F}^n$. We say “ $(x_1, \dots, x_n)$ is transferable to $(y_1, \dots, y_n)$ using σ ” or write “ $(x_1, \dots, x_n) \xrightarrow{\sigma} (y_1, \dots, y_n)$ ” if there exists a sequential addition $f : \mathbb{F} \rightarrow \mathbb{F}$ using σ such that $f(x_i) = y_i$ for all $i \in [n]$.

The following result is a powerful tool for characterizing the representational power of sequential additions.

Lemma A.3 (Lemma 4.6 of Hwang et al. (2025b)). Let $\sigma : \overline{\mathbb{F}} \rightarrow \overline{\mathbb{F}}$ and suppose that σ satisfies Condition 1. Then, for any $y \in [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{F}}$ and $x_1, x_2 \in [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{F}}$ such that $x_2 - x_1 \in (0, 2^{\epsilon_{\max}}]_{\mathbb{F}}$, it holds that

$$(-2^{\epsilon_{\max}}, y, y^+, 2^{\epsilon_{\max}}) \xrightarrow{\sigma} (x_1, x_1, x_2, x_2).$$

Note that Condition 1 in Hwang et al. (2025b) is replaced here by Condition 1. Since any activation function satisfying Condition 1 also satisfies Condition 1 of Hwang et al. (2025b), this replacement is valid.

Corollary A.4. Let $\sigma : \overline{\mathbb{F}} \rightarrow \overline{\mathbb{F}}$ and suppose that σ satisfies Condition 1. Then, for any $x, y \in \mathbb{F}$,

$$(x) \xrightarrow{\sigma} (y).$$

Proof. By Lemma A.3, we get the desired result for $|x|, |y| \leq 2^{\epsilon_{\max}}$. Therefore, it is sufficient to prove that

$$(x) \xrightarrow{\sigma} (2^{\epsilon_{\max}}), \text{ and } (2^{\epsilon_{\max}}) \xrightarrow{\sigma} (y),$$

for $|x|, |y| > 2^{\epsilon_{\max}}$.

First suppose $x, y > 2^{\epsilon_{\max}}$. By the second item of Condition 1, we have $[\epsilon_{\min} - M, \epsilon_{\max} - M]_{\mathbb{Z}} \subset \bigcup_{i=1}^{\nu} [\epsilon_{\sigma(\gamma_i)} + \epsilon_{\min}, \epsilon_{\max} - \kappa + \min(\epsilon_{\sigma(\gamma_i)}, k_i)]_{\mathbb{Z}}$. Hence there exists γ_i such that $\epsilon_{\max} - M \leq \epsilon_{\max} - \kappa + \min(\epsilon_{\sigma(\gamma_i)}, k_i)$ leading to

$\min(\epsilon_{\sigma(\gamma_i)}, k_i) \geq -M$. and $2^{-M} \leq |\sigma(\gamma_i)|$. Therefore, there exists $w \in \mathbb{F}$ such that

$$\frac{1}{2} \times 2^{\epsilon_{\max} - M} < w \otimes \sigma(\gamma_i) < \frac{3}{2} \times 2^{\epsilon_{\max} - M}.$$

Then, since $z \oplus ((-w) \otimes \sigma(\gamma_i)) = z^-$ for $z > 2^{\epsilon_{\max}}$ and $z \oplus (w \otimes \sigma(\gamma_i)) = z^+$ for $z \geq 2^{\epsilon_{\max}}$, there exists $n_x, n_y \in \mathbb{N}$ such that

$$\begin{aligned} x \oplus \bigoplus_{j=1}^{n_x} (-w) \otimes \sigma(\gamma_i) &= 2^{\epsilon_{\max}}, \\ 2^{\epsilon_{\max}} \oplus \bigoplus_{j=1}^{n_y} (w \otimes \sigma(\gamma_i)) &= y. \end{aligned}$$

Similar arguments apply to other cases. This completes the proof. $\square$

Definition A.5. Now let $\alpha_1, \alpha_2, w \in \mathbb{F}$ such that

$$\alpha_1 \otimes w, \alpha_2 \otimes w \in \mathbb{F}.$$

We define $A_{(\alpha_1), (\alpha_2), w}$ as the set of affine transformations with sequential additions

$$A_{(\alpha_1), (\alpha_2), w} := \{x \mapsto s(w \otimes x) : \mathbb{F} \rightarrow \mathbb{F} \mid s : (w \otimes \alpha_1) \xrightarrow{\sigma} (\alpha_2)\}.$$

Let $\alpha_1, \beta_2, \alpha_2, \beta_2, w \in \mathbb{F}$ such that

$$\alpha_1 \otimes w \neq \beta_1 \otimes w,$$

and

$$|\alpha_1 \otimes w|, |\beta_1 \otimes w|, |\alpha_2|, |\beta_2|, |\alpha_2 - \beta_2| \leq 2^{\epsilon_{\max}}.$$

If $w(\alpha_1 - \beta_1)(\alpha_2 - \beta_2) > 0$, we define $A_{(\alpha_1, \beta_1), (\alpha_2, \beta_2), w} : \mathbb{F} \rightarrow \mathbb{F}$ as the set of affine transformations with sequential additions

$$A_{(\alpha_1, \beta_1), (\alpha_2, \beta_2), w} := \{x \mapsto s(w \otimes x) : \mathbb{F} \rightarrow \mathbb{F} \mid s : (w \otimes \alpha_1, w \otimes \beta_1) \xrightarrow{\sigma} (\alpha_2, \beta_2)\}.$$

If $w(\alpha_1 - \beta_2)(\alpha_2 - \beta_2) < 0$, $A_{(\alpha_1, \beta_1), (\alpha_2, \beta_2), w}$ is defined as

$$A_{(\alpha_1, \beta_1), (\alpha_2, \beta_2), w} := A_{(\alpha, \beta), (\alpha_2, \beta_2), -w}.$$

Note that, by Lemma A.3, such a sequential addition exists since $|\sigma(\alpha_1) \otimes w|, |\sigma(\beta_1) \otimes w|, |\alpha_2|, |\beta_2|, |\alpha_2 - \beta_2| \leq 2^{\epsilon_{\max}}$.

One of our key ideas is to utilize the following module to control the input gradient while preserving the function value.

Definition A.6. Let $\alpha_1, \alpha_2, w \in \mathbb{F}$ or $\alpha_1, \beta_1, \alpha_2, \beta_2, w \in \mathbb{F}$ satisfy the same conditions as in Definition A.5. We define $A_{(\alpha_1), (\alpha_2), w}$ and $AS_{(\alpha_1, \beta_1), (\alpha_2, \beta_2), w}$ as follows:

$$AS_{(\alpha_1), (\alpha_2), w} := \{f \circ \sigma : \mathbb{F} \rightarrow \mathbb{F} \mid f \in A_{(\sigma(\alpha_1)), (\alpha_2), w}\},$$

and

$$AS_{(\alpha_1, \beta_1), (\alpha_2, \beta_2), w} := \{f \circ \sigma : \mathbb{F} \rightarrow \mathbb{F} \mid f \in A_{(\sigma(\alpha_1), \sigma(\beta_1)), (\alpha_2, \beta_2), w}\}.$$

Note that for any $f \in AS_{(\alpha_1, \beta_1), (\alpha_2, \beta_2), w}$, we have

$$f(x) = \begin{cases} \alpha_2 & \text{if } x = \alpha_1, \\ \beta_2 & \text{if } x = \beta_1. \end{cases}$$

Moreover,

$$\mathcal{D}_{f,x}(g) = \begin{cases} g \otimes w \otimes \sigma'(\alpha_1) & \text{if } x = \alpha_1, \\ g \otimes w \otimes \sigma'(\beta_1) & \text{if } x = \beta_1. \end{cases}$$

Lemma A.7. Let $\sigma : \mathbb{F} \rightarrow \mathbb{F}$ and suppose that σ satisfies Condition 1. For $\alpha_1, \alpha_2, w \in \mathbb{F}$ with $w \otimes \alpha_1, w \otimes \alpha_2 \in \mathbb{F}$, then $A_{(\alpha_1), (\alpha_2), w} \neq \emptyset$. i.e. such sequential addition always exists. Moreover, if $w \otimes \sigma(\alpha_1), w \otimes \sigma(\alpha_2) \in \mathbb{F}$, then $AS_{(\alpha_1), (\alpha_2), w} \neq \emptyset$.

Proof. It follows from Corollary A.4. □

Definition A.8. We define $\langle AS \rangle_{(\alpha_{\text{in}}), (\alpha_{\text{out}})}$ as follows:

$$\langle AS \rangle_{(\alpha_{\text{in}}), (\alpha_{\text{out}})}^n := \{f = f_n \circ \dots \circ f_1 : \mathbb{F} \rightarrow \mathbb{F} \mid f_i = AS_{(\alpha_{i-1}), (\alpha_i), w_i}, \alpha_0 = \alpha_{\text{in}}, \alpha_n = \alpha_{\text{out}}, i \in [n]\}.$$

We define $\langle AS \rangle_{(\alpha_{\text{in}}, \beta_{\text{in}}), (\alpha_{\text{out}}, \beta_{\text{out}})}$ as follows:

$$\begin{aligned} \langle AS \rangle_{(\alpha_{\text{in}}, \beta_{\text{in}}), (\alpha_{\text{out}}, \beta_{\text{out}})}^n := \{f = f_n \circ \dots \circ f_1 : \mathbb{F} \rightarrow \mathbb{F} \mid f_i = AS_{(\alpha_{i-1}, \beta_{i-1}), (\alpha_i, \beta_i), w_i}, \alpha_0 = \alpha_{\text{in}}, \beta_0 = \beta_{\text{in}}, \\ \alpha_n = \alpha_{\text{out}}, \beta_n = \beta_{\text{out}}, i \in [n]\}. \end{aligned}$$

In other words, $\langle AS \rangle_{(\alpha_{\text{in}}), (\alpha_{\text{out}})}^n$ and $\langle AS \rangle_{(\alpha_{\text{in}}, \beta_{\text{in}}), (\alpha_{\text{out}}, \beta_{\text{out}})}^n$ are the set of n -layered neural networks that start with an activation layer and map $(\alpha_{\text{in}},)$ to $(\alpha_{\text{out}},)$ and $(\alpha_{\text{in}}, \beta_{\text{in}})$ to $(\alpha_{\text{out}}, \beta_{\text{out}})$, respectively.

Definition A.9 (Gradient Transferability). Assume that σ satisfies Condition 2. We write

$$(g_1, \dots, g_m) \xrightarrow{n} (g'_1, \dots, g'_m),$$

if there exists $f \in \langle AS \rangle_{(\alpha_1), (\alpha_2)}^n$ such that

$$\mathcal{D}_{f, \alpha_1}(g_i) = g'_i.$$

For a scalar value, we denote $g \xrightarrow{n} g_2$ if and only if $(g) \xrightarrow{n} (g_2)$. When the precise target value is not important—except that it does not overflow and is distinct from other outputs—we use the symbol “−”. For example,

$$(g_1, g_2) \xrightarrow{n} (g'_1, -),$$

means that there exists $f \in \langle AS \rangle_{(\alpha_1), (\alpha_2)}^n$ such that

$$\mathcal{D}_{f, \alpha_1}(g_1) = g'_1,$$

and

$$g'_1 \neq \mathcal{D}_{f, \alpha_1}(g_2) \in \mathbb{F}.$$

Definition A.10 (Separating Point, Definition 3.5 in (Hwang et al., 2025b)). For $\sigma : \mathbb{F} \rightarrow \mathbb{F}$, we say that $\theta \in \mathbb{F}$ is a “separating point of σ ” if

$$\sigma(\theta^-) \notin \{\sigma(\theta), \sigma(\theta^+)\} \quad \text{or} \quad \sigma(\theta^+) \notin \{\sigma(\theta), \sigma(\theta^-)\}.$$

Definition A.11 (Activation functions). We present the definitions of widely used activation functions.

- Sigmoid(x) := $\frac{1}{1+e^{-x}}$,
- ReLU(x) := $\max(0, x)$,
- ELU(x) := $\begin{cases} x & \text{if } x \geq 0, \\ \exp(x) - 1 & \text{if } x < 0, \end{cases}$
- SiLU(x) := $\frac{x}{1+e^{-x}}$,
- SoftPlus(x) := $\log(1 + \exp(x))$,
- GELU(x) := $\frac{x}{2} \left(1 + \operatorname{erf} \left(\frac{x}{\sqrt{2}} \right) \right) = \frac{x}{2} \left(1 + \frac{2}{\sqrt{\pi}} \int_0^{x/\sqrt{2}} e^{-t^2} dt \right)$,

We also present the specification of various floating-point formats in Table 2.

Table 2. List of frequently-used floating-point formats.

Floating-point format	(M, E)
16-bit half-precision (IEEE, 2019)	(10, 5)
32-bit single-precision (IEEE, 2019)	(23, 8)
64-bit double-precision (IEEE, 2019)	(52, 11)
8-bit E5M2 (Micikevicius et al., 2022)	(2, 5)
8-bit E4M3 (Micikevicius et al., 2022)	(3, 4)
bfloat16 (Google; Abadi et al., 2016)	(7, 8)

B. Lemmas and Proofs for General Activation Functions

The following lemma is used to align the number of layers in networks of the form $AS_{(\delta_0, \delta_1), (\delta_0, \delta_1), w}$.

Lemma B.1. *Suppose σ satisfies Condition 2. Then there exists a floating-point number $w \in \mathbb{F}$ and a neural network $f \in AS_{(\delta_0, \delta_1), (\delta_0, \delta_1), w}$ such that, for any $g \in \mathbb{F}$ satisfying $|g| \leq \Omega \times 2^{\epsilon \sigma'(\delta_1)}$,*

$$\mathcal{D}_{f, \delta_1}(g) = g,$$

and

$$|\mathcal{D}_{f, \delta_0}(g)| \leq |g|.$$

Proof. Take $w := 2^{-\epsilon \sigma'(\delta_1)}$. Since $|\mathcal{D}_{f, x}(g)| = g \otimes w \otimes \sigma'(x)$, the claim then follows directly from the definition. $\square$

B.1. Lemmas about Condition 1

Lemma B.2. *Let $K \in \mathbb{F}$ with $|K| \in [(1 + 2^{-M+1}) \times 2^{-M-2}, 1 + 2^{-2} - 2^{-M}]_{\mathbb{F}}$. Consider $\epsilon_\zeta \in \mathbb{Z}$ such that $\epsilon_{\min} - M \leq \epsilon_\zeta \leq \epsilon_{\max} - M$. Then, there exists $\gamma \in \mathbb{F}$ such that the following inequality holds:*

$$\frac{1}{2} \times 2^{\epsilon_\zeta} < \gamma \otimes K \leq \frac{5}{4} \times 2^{\epsilon_\zeta}.$$

Furthermore, $\gamma \leq 2^{\epsilon_\zeta - \epsilon_K}$.

Proof. This result follows from Lemma 24 of (Hwang et al., 2025a). The range of γ follows directly from the proof. $\square$

B.2. Proof of Lemma 4.1

Proof. Lemma 4.1 is a direct corollary of Lemma 4.1 in (Hwang et al., 2025b), which constructs a network $\tilde{f}$ such that $\tilde{f}(x) = \sigma(c) \mathbb{1}_z(x)$. Note that $\tilde{f}$ can be decomposed as $\tilde{f} = \sigma \circ \rho \circ f_0$, where $n \in \mathbb{N}$, $f_0 : \mathbb{F}^d \rightarrow \mathbb{F}^n$ is a two-layer neural network, and $\rho : \mathbb{F}^n \rightarrow \mathbb{F}$ is an affine transformation whose weights are all 0, 1, or -1 . Write $\rho(x) = W \otimes x \oplus b$.

Define $\rho_2 : \begin{bmatrix} W \\ W \end{bmatrix} \otimes x \oplus \begin{bmatrix} b \\ b \end{bmatrix}$. Define $f : \mathbb{F}^d \rightarrow \mathbb{F}^2$ as $f := \sigma \circ \rho_2 \circ f_0$. Then, $f = (\tilde{f}, \tilde{f})$ and $\mathcal{D}_{f, x} \begin{bmatrix} g & -g \end{bmatrix} = \mathcal{D}_{f_0, x}(\mathcal{D}_{\sigma \circ \rho, f_0(x)}) \begin{bmatrix} g & -g \end{bmatrix} = 0$. This completes the proof. $\square$

B.3. Proof of Lemma 4.2

Proof. Without loss of generality, assume that $(\sigma(\delta_1) - \sigma(\delta_0))(y_2 - y_1) \geq 0$. In other case, we can flip the sign of weights.

Define $f : \mathbb{F}^2 \rightarrow \mathbb{F}$ as follows:

$$f(x_1, x_2) := \sigma(s_2(s_1(x_1) \oplus (-x_2))),$$

where s_1, s_2 are sequential additions (Lemma A.3) satisfying

$$s_1 : (\sigma(\delta_0), \sigma(\delta_1)) \xrightarrow{\sigma} (2\sigma(\delta_0), 2\sigma(\delta_0)),$$

and

$$s_2 : (\sigma(\delta_0), \sigma(\delta_1)) \xrightarrow{\sigma} (y_1, y_2).$$

This construction satisfies all assumptions of the lemma. This completes the proof. $\square$

B.4. Proof of Lemma 3.7

Once Lemma 3.7 is established, taking f_1 from Lemma 3.4 and f_3 from Lemma 3.7, the sum $f_1 \# f_3$ completes the proof Theorem 3.8. Therefore, it suffices to prove Lemma 3.7.

Proof of Lemma 3.7. The proof follows the outline of Lemma 3.5. Unlike in Lemma 3.5, the input gradient is independent of the input value, which necessitates the construction of a gradient indicator function depending solely on the gradient value. Lemma B.3 establishes this construction.

Let $\{t_1 < \dots < t_l\} = [0, \Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}]_{\mathbb{F}}$. As before, we use networks $\psi_{i,21}$ and $\psi_{i,22}$ defined in the proof of Lemma 3.5 to construct the gradient indicator function. The primary difference lies in replacing $\psi_{i,22}$ with the composition $\psi_{i,23,j} \circ \psi_{i,22}$ where $\psi_{i,23,j}$ is a neural network from Lemma B.3: $\psi_{i,23,j} : \delta_1 \mapsto 0$ and

$$\mathcal{D}_{\psi_{i,23,j}} = \begin{cases} 0 & \text{if } 0 \leq |g| < t_j, \\ \omega & \text{if } t_j \leq g \leq \Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}, \\ -\omega & \text{if } -t_j \geq g \geq -\Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}. \end{cases} \quad (8)$$

Then we obtain

$$\mathcal{D}_{\psi_{i,23,j} \circ \psi_{i,22}, \delta_1}(g) = \begin{cases} \omega & \text{if } t_j \leq g \leq \Omega \times 2^{\epsilon_{\sigma'}(\delta_1)} \text{ and } x = z_i, \\ -\omega & \text{if } -t_j \geq g \geq -\Omega \times 2^{\epsilon_{\sigma'}(\delta_1)} \text{ and } x = z_i, \\ 0 & \text{if } 0 \leq |g| < t_j \text{ or } x \neq z_i. \end{cases}$$

Then, using arguments similar to those in the proof of Lemma 3.5, we obtain $\phi_{i,j}$ such that $\phi_{i,j}(x) = 0$ and

$$\mathcal{D}_{\phi_{i,j},x}(y) = \begin{cases} g^*(x, y) \otimes \mathbb{1}_{x=z_i, y \geq t_j} & \text{if } y \geq 0, \\ g^*(x, y) \otimes \mathbb{1}_{x=z_i, y \leq -t_j} & \text{if } y < 0. \end{cases}$$

Define Φ_i as

$$\Phi_i := \#_{j \in [m_2]} \phi_{i,j} \ominus \phi_{i,j+1},$$

where $\phi_{i,m_2+1} := 0$. One can verify that $\Phi_i(x) = 0$ and

$$\mathcal{D}_{\Phi_i,x}(y) = \begin{cases} g^*(z_i, y) & \text{if } x = z_i, \\ 0 & \text{if } x \neq z_i. \end{cases}$$

$f := \#_{i=1}^m \Phi_i$ gives the desired result. □

Lemma B.3. Suppose σ satisfies Condition 2. Let $g_0 \in \mathbb{F}$ such that $g_0 > 0$. Then, for $L \geq 2^{E+1} + 2M + 2$, there exists an L -layer neural network $f : \mathbb{F} \rightarrow \mathbb{F}$ such that $f(\delta_1) = 0$, and

$$\mathcal{D}_{f,\delta_1}(g) = \begin{cases} 0 & \text{if } 0 \leq |g| < g_0, \\ \omega & \text{if } g_0 \leq g \leq \Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}, \\ -\omega & \text{if } -g_0 \geq g \geq -\Omega \times 2^{\epsilon_{\sigma'}(\delta_1)}. \end{cases}$$

Proof. As the gradient is symmetric with respect to zero, it suffices to consider nonnegative inputs. Define u as $u := 2^{\epsilon_{\max} + \epsilon_{\sigma'}(\delta_1)}$. It is sufficient to prove that for $T = 2^{E+1} + 2M + 2$,

$$(0, g_0^-, g_0, u) \xrightarrow{T} (0, 0, \omega, \omega).$$

Suppose that $2^i \omega \leq g_0 < 2^{i+1} \omega$ for some $i \in \mathbb{Z}$ with $0 \leq i < M - 1$. Then, by Lemma B.4, we obtain

$$(0, g_0^-, g_0, 2^{\epsilon_{\max} - M + \epsilon_{\sigma'}(\delta_1)}) \xrightarrow{1} (0, -, (2^i + 1) \omega, u'),$$

where $u' < u$. We use $h \in AS_{(\delta_1, \cdot), (\delta_1, \cdot), w}$, where $w = (1 + 2^{-M}) \times 2^{-i-1-\epsilon_{\sigma'(\delta_1)}}$. Then because $\epsilon_{\sigma'(\delta_1)} \leq 0$

$$(2^i \omega) \otimes w = \left[(2^{-\epsilon_{\sigma'(\delta_1)}-1} + 2^{-\epsilon_{\sigma'(\delta_1)}-1-M}) \times \omega \right] = 2^{-\epsilon_{\sigma'(\delta_1)}-1} \omega < ((2^i + 1) \omega) \otimes w.$$

Therefore,

$$\mathcal{D}_{h, \delta_1}(g) = g \otimes w \otimes \sigma'(\delta_1) = \begin{cases} 0 & \text{if } g = 2^i \omega, \\ v_1 & \text{if } g = (2^i + 1) \omega, \\ v_2 & \text{if } g > 2^i \omega, \end{cases}$$

where $v_1 > 0$ and $g > v_2 \in \mathbb{F}$. Thus, it suffices to prove that

$$(0, \omega, u) \xrightarrow{2^{E+1}+2M-2} (0, \omega, \omega).$$

Use $h_2 \in AS_{(\delta_1, \cdot), (\delta_1, \cdot), w_2}$ where $w_2 = 1^+ \otimes 2^{-\epsilon_{\sigma'(\delta_1)}-1}$. Then for any $g \in \mathbb{F}$ with $\omega \leq g \leq u$,

$$\mathcal{D}_{h_2, \delta_1}(g) = g \otimes w_2 \otimes \sigma'(\delta_1) = \lceil 1^+ \times (\mathbf{m}_g \times 2^{\epsilon_g}) \times 2^{-1} \rceil \leq \lceil (\mathbf{m}_g + 2^{-M} \times \mathbf{m}_g) \times 2^{\epsilon_g-1} \rceil \leq g^{++} \otimes 2^{-1}.$$

And $\mathcal{D}_{h_2^{\circ 2}, \delta_1}(g) \leq g^{++++} \times 2^{-2} \leq \frac{g}{2}$ since $M \geq 2$. By applying h_2 a total of $2^{E+1} + 2M - 2$ times, we get

$$(0, \omega, u) \xrightarrow{2^{E+1}+2M-2} (0, \omega, \omega).$$

Now consider the case $g_0 \geq 2^{\epsilon_{\min}}$. By Lemma B.5, there exist $x_1, x_2, x_3 \in (1/2, 1]_{\mathbb{F}}$ such that

$$\mathbf{m}_{g_0^-} \otimes x_1 \otimes x_2 \otimes x_3 = 1,$$

and

$$\mathbf{m}_{g_0} \otimes x_1 \otimes x_2 \otimes x_3 = 1^+.$$

Define $w_i \in \mathbb{F}$ for $i = 1, 2, 3$ by $w_i := x_i \times 2^{-\epsilon_{\sigma'(\delta_1)}-n_i}$, where $n_1, n_2, n_3 \in [\epsilon_{\min}, \epsilon_{\max}]_{\mathbb{N}_0}$ satisfy $\epsilon_{g_0} - \epsilon_{\min} = n_1 + n_2 + n_3$, and let $f_i \in AS_{(\delta_1, \cdot), (\delta_1, \cdot), w_i}$. Then

$$(g_0^-, g_0, u) \xrightarrow{3} (-, g_0 \otimes w_1 \otimes w_2 \otimes w_3, u_2) = (-, 1^+ \times 2^{\epsilon_{\min}}, u_2),$$

for some $u_2 \in \mathbb{F}$ with $0 < u_2 < u$. By using $h_4 \in AS_{(\delta_1, \cdot), (\delta_1, \cdot), w_4}$, where

$$w_4 = \begin{cases} (1 + 2^{\epsilon_{\sigma'(\delta_1)}}) \times 2^{-\epsilon_{\sigma'(\delta_1)}-M-1} & \text{if } \epsilon_{\sigma'(\delta_1)} < -1, \\ (1 + 2^{-1})^- \times 2^{-M} & \text{if } \epsilon_{\sigma'(\delta_1)} = -1, \\ 2^{-M-1} & \text{if } \epsilon_{\sigma'(\delta_1)} = 0, \end{cases}$$

we get

$$(-, 1^+ \times 2^{\epsilon_{\min}}, u_2) \xrightarrow{1} (0, \omega, u_3),$$

for some $u_3 \in \mathbb{F}$ with $0 < u_3 < u_2$. Applying the same argument as above, we conclude that

$$(0, \omega, u_3) \xrightarrow{2^{E+1}+2M-2} (0, \omega, \omega).$$

This completes the proof. □

Lemma B.4. Consider $-k \in [-M, 0]_{\mathbb{Z}}$. For every $i \in [0, M]_{\mathbb{Z}}$ and $g \in [2^i \omega, 2^{i+1} \omega)$, there exists $x \in (2^{k-1}, 2^k]_{\mathbb{F}}$ such that

$$g \otimes x \leq 2^i \omega,$$

and

$$g^+ \otimes x = (2^i + 1) \omega.$$

Proof. Let $\mathcal{M} = \{1 + \sum_{i=1}^M b_i 2^{-i} : b_1, \dots, b_n \in \{0, 1\}\} \cup \{2\}$. Then, there exists $\alpha, \alpha' \in \mathcal{M}$ such that $g = \alpha \times 2^i$, $g^+ = \alpha' \times 2^i$, and $\alpha' - \alpha = 2^{-i}$. Let d be the largest number in $\mathcal{M}$ such that $\alpha \times (d/2) \leq (1 + 2^{-i-1})$. Then, it is easy to observe that such d always exists and $1 < d < 2$. Furthermore, since $\alpha < 2$, $d < 2$, and $d^+ - d \leq 2^{-M}$, it holds that

$$(1 + 2^{-i-1}) - \alpha \times (d/2) < \alpha \times (d^+/2) - \alpha \times (d/2) < 2^{-M}.$$

This implies that $\alpha \times (d/2) > 1 + 2^{-i-1} - 2^{-M}$, and hence,

$$\alpha' \times (d/2) = \alpha \times (d/2) + (d/2) \times 2^{-i} > \alpha \times (d/2) + 2^{-i-1} > 1 + 2^{-i} - 2^{-M}$$

as $d > 1$. Let $x = d \times 2^{k-1}$. Then, $g \otimes x \in [2^{i+k}, (1 + 2^{-i-1}) \times 2^{i+k}]_{\mathbb{F}}$ and $g^+ \otimes x \in ((1 + 2^{-i-1}) \times 2^{i+k}, (1 + 2^{-i}) \times 2^{i+k}]_{\mathbb{F}}$. This implies $g \otimes x \otimes 2^{-k} = 2^i \omega$ and $g^+ \otimes x \otimes 2^{-k} = (1 + 2^i) \times \omega$, and completes the proof. $\square$

Lemma B.5 (To $(1^+, 1)$). *For every $g \in [1, 2]_{\mathbb{F}}$, there exist $x_1, x_2, x_3 \in (1/2, 1]_{\mathbb{F}}$ such that*

$$g \otimes x_1 \otimes x_2 \otimes x_3 = 1,$$

and

$$g^+ \otimes x_1 \otimes x_2 \otimes x_3 = 1^+.$$

Proof. If $g = 1$, it is trivial. Let $h = g^+$, $h^- = g$. Denote $a^\dagger \in \mathbb{F}$ such that $a \otimes a^\dagger = 1^+$. By Lemma B.7, the such $x^\dagger$ always exists. Let $h = 1 + n2^{-M}$. Then we have

$$1 > (1 + n \cdot 2^{-M})(1^+ - n \cdot 2^{-M}) = 1^+ - n^2 2^{-2M}.$$

First note that for $a, b \in \mathbb{F}$ with $a \in (1, 2)_{\mathbb{F}}$, $b \in (2^{-1}, 1)_{\mathbb{F}}$, if $r = ab - (a \otimes b) < 0$, we have

$$a^- \otimes b < a \times b,$$

since

$$a^- b = (a - 2^{-M})b = ab - 2^{-M}b < a \otimes b - 2^{-M}b < a \otimes b - 2^{-M-1}.$$

Let $\mathcal{R}_0 := \{1^+ + n \cdot 2^{-M} : n \leq 2^{\frac{M-1}{2}}, n \in \mathbb{N}\}$. If $n < 2^{\frac{M-1}{2}}$, since

$$1 > (1^+ + n \cdot 2^{-M})(1^+ - n \cdot 2^{-M}) = 1^+ - n^2 2^{-2M} > 1^+ - 2^{-M-1},$$

we have $(1^+ + n \cdot 2^{-M}) \otimes (1^+ - n \cdot 2^{-M}) = 1^+$. and $(1^+ + (n-1) \cdot 2^{-M}) \otimes (1^+ - n \cdot 2^{-M}) = 1$.

If $n = 2^{\frac{M-1}{2}}$, since

$$(1^+ + n \cdot 2^{-M})(1^+ - n \cdot 2^{-M} + 2^{-M-1}) = 1^+ + (n+1)2^{-2M-1} = 1^+ + 2^{-\frac{3}{2}M-\frac{3}{2}} + 2^{-2M-1},$$

$$(1^+ + n \cdot 2^{-M} + 2^{-M})(1^+ - n \cdot 2^{-M} + 2^{-M-1}) = 1^+ + 2^{-\frac{3}{2}M-\frac{3}{2}} + n2^{-2M} - 2^{-2M} = 1 + 3 \cdot 2^{-\frac{3}{2}M-\frac{3}{2}} - 2^{-2M} < 1 + 2^{-M-1},$$

we have

$$(1^+ + n \cdot 2^{-M}) \otimes (1^+ - n \cdot 2^{-M} + 2^{-M-1}) = 1^+, (1^+ + n \cdot 2^{-M} + 2^{-M}) \otimes (1^+ - n \cdot 2^{-M} + 2^{-M-1}) = 1.$$

Therefore, if $a \in \mathcal{R}_0$ with $a \otimes x = 1^+$, then $a^- \otimes x = 1$.

Let $x_h \in (2^{-1}, 1)_{\mathbb{F}}$ such that $h \otimes x_h = 1^+$ and $r_h = 1^+ - hx_h$. If $r_h \leq 0$, then the proof is finished with $x_1 = x_h$. Suppose $r_h > 0$. We find the smallest $m \in \mathbb{N}$, such that $hx_h^{m+} - h \otimes x_h^{m+} \leq 0$. Note that if $r_h > 0$, then $h \otimes x_h^+ > h \otimes x_h$ since $h \times x_h^+ = 1^+ + r_h + h2^{-M-1} > 1 + 2^{-M-1}$. Therefore if $hx_h^{m+} - h \otimes x_h^{m+} > 0$, we have $h \otimes x_h^{m+} = 1^+ + m2^{-M}$. Hence

$$hx_h^{m+} - h \otimes x_h^{m+} = h(x_h + m2^{-M-1}) - (1^+ + m2^{-M}) = r_h - m(2 - h)2^{-M-1} > 0.$$

Therefore, we have

$$m < \frac{r_h}{(2-h)2^{-M-1}} \leq \frac{2^{-M-1}}{(2-h)2^{-M-1}} = \frac{1}{2-h},$$

leading to $m_h \geq \frac{1}{2-h}$.

We consider the following cases.

- **Case (1)** $h \leq 3 \times 2^{-1}$.

In this case, we have $m_h \leq 2$ and $h \times x_h^{m_h+} \leq hx_h + h(m_h 2^{-M-1}) \leq 1^{++} + 3 \cdot 2^{-M-1} \leq 1^{4+}$. Hence $h \otimes x_h^{m_h+} \in \mathcal{R}_0$, and it completes the proof with $x_1 = x_h^{m_h+}$, $x_2 = (h \otimes x_h^{m_h+})^\dagger$.

- **Case (2)** $3 \times 2^{-1} < h \leq 2^{3-}$.

In this case, we have $m_h \leq \frac{1}{2-h} \leq \frac{2^M}{3}$ and $h \times x_h^{m_h+} \leq hx_h + h(m_h 2^{-M-1}) \leq (1 + 3 \cdot 2^{-M-1}) + \frac{h}{6} \leq 3 \times 2^{-1}$. Hence it reduces to case (1).

- **Case (3)** $h = 2^{-}$.

In this case, we have

$$\begin{aligned} (2 - 2 \cdot 2^{-M})(2^{-1} + 2^{-2} + 2^{-M}) &= 1 + 2^{-1} + 2^{-M-1} - 2^{-2M+1}, & (2 - 2 \cdot 2^{-M}) \otimes (2^{-1} + 2^{-2} + 2^{-M}) &= 1 + 2^{-1}, \\ (2 - 3 \cdot 2^{-M})(2^{-1} + 2^{-2} + 2^{-M}) &= 1 + 2^{-1} - 2^{-M-1} - 3 \cdot 2^{-2M}, & (2 - 3 \cdot 2^{-M}) \otimes (2^{-1} + 2^{-2} + 2^{-M}) &= 1 + 2^{-1} - 2^{-M}. \end{aligned}$$

Hence it reduces to case (2) with $x_1 = 2^{-1} + 2^{-2} + 2^{-M}$, $h \otimes x_1 = 1 + 2^{-1}$, $h^- \otimes x_1 = 1 + 2^{-1} - 2^{-M}$.

- **Case (4)** $h = 2^{-}$.

In this case, we have

$$\begin{aligned} (2 - 2^{-M})(2^{-1} + 2^{-2}) &= 1 + 2^{-1} - 3 \cdot 2^{-M-2}, & (2 - 2^{-M}) \otimes (2^{-1} + 2^{-2}) &= 1 + 2^{-1} - 2^{-M}, \\ (2 - 2 \cdot 2^{-M})(2^{-1} + 2^{-2}) &= 1 + 2^{-1} - 3 \cdot 2^{-M-1}, & (2 - 2 \cdot 2^{-M}) \otimes (2^{-1} + 2^{-2}) &= 1 + 2^{-1} - 2 \cdot 2^{-M}. \end{aligned}$$

Hence it reduces to case (2) with $x_1 = 2^{-1} + 2^{-2}$, $h \otimes x_1 = 1 + 2^{-1} - 2^{-M}$, $h^- \otimes x_1 = 1 + 2^{-1} - 2 \cdot 2^{-M}$. □

Lemma B.6. For any $x \in \mathbb{F}$ and $2^e \in \mathbb{F}$ for some $e \in \mathbb{N}$, we have $x \otimes \underbrace{2^e \otimes \dots \otimes 2^e}_{m \text{ times}} = 0$ for $m = \left\lceil \frac{2^E + M}{e} \right\rceil_{\mathbb{Z}}$.

Proof. Since $|x| < 2^{1+\epsilon_{\max}}$, we have

$$\begin{aligned} |x \otimes \underbrace{2^e \otimes \dots \otimes 2^e}_{m \text{ times}}| &\leq 2^{1+\epsilon_{\max}} \otimes \underbrace{2^e \otimes \dots \otimes 2^e}_{m \text{ times}} \\ &= 2^{1+\epsilon_{\max}-em} = 2^{1+2^{E-1}-(2^E+M)} = 2^{1-2^{E-1}-M} = 2^{\epsilon_{\min}-M-1} = \frac{1}{2}\omega. \end{aligned}$$

Hence $|x \otimes \underbrace{2^e \otimes \dots \otimes 2^e}_{m \text{ times}}| = 0$. □

Lemma B.7 (Lemma 17 of [Park et al. \(2026\)](#)). Suppose $M \geq 2$. For any $x \in (1, 2]_{\mathbb{F}}$, there exist $y \in (2^{-1}, 1]_{\mathbb{F}}$ and $z \in (2^{-1}, 1^+]_{\mathbb{F}}$ such that $x \otimes y = 1^+$.

B.5. Proof of Lemma 4.3

Proof. By Lemma B.9, it suffices to show that there exist $w \in \mathbb{F}$ and $\gamma \in \{\gamma_0, \dots, \gamma_\nu\}$ such that

$$\frac{1}{2} \times 2^{\epsilon_c - M} < (w \otimes \sigma(\gamma_i)) < \frac{3}{2} \times 2^{\epsilon_c - M}, \quad \epsilon_c \in [\epsilon_{\min} - M, \epsilon_{\max} - M]_{\mathbb{Z}}$$

and

$$g \otimes w \otimes \sigma'(\gamma_i), \quad g \otimes w \otimes \sigma'(\gamma_0) \in \mathbb{F}.$$

By Condition 1, there exists $i \in [\nu]$ such that

$$\mathbf{e}_c \in [\mathbf{e}_{\sigma(\gamma_i)} + \mathbf{e}_{\min}, \mathbf{e}_{\max} - \kappa + \min(\mathbf{e}_{\sigma(\gamma_i)}, k_i)].$$

By Lemma B.2, there exists $v \in \mathbb{F}$ such that $\frac{1}{2} < v \otimes \mathbf{m}_{\sigma(\gamma_i)} < \frac{3}{2}$. Let $w = e^{\mathbf{e}_c - \mathbf{e}_{\sigma(\gamma_i)} - M} \times v$ and we have

$$\frac{1}{2} \times 2^{\mathbf{e}_c - M} < w \otimes \sigma(\gamma_i) = v \otimes 2^{\mathbf{e}_c} < \frac{3}{2} \times 2^{\mathbf{e}_c - M}.$$

Moreover,

$$|g \otimes w \otimes \sigma'(\gamma_i)| \leq 2^\kappa \otimes w \otimes 2^{-k_i} |\sigma(\gamma_i)| \leq \frac{3}{2} \times 2^{\mathbf{e}_c - M + \kappa - k_i} \leq \frac{3}{2} \times 2^{\mathbf{e}_c} < \Omega.$$

A similar estimate shows that

$$|g \otimes w \otimes \sigma'(\gamma_0)| \leq \Omega.$$

This completes the proof. $\square$

B.6. Proof of Lemma 4.4

Lemma B.8. *Suppose σ satisfies Condition 2. Consider $g_1, g_0 \in \mathbb{F}$ such that $0 < |g_1| \leq \Omega \times 2^{\mathbf{e}_{\sigma'(\delta_1)}}$. Then, for any $L \geq \left\lceil \max\left(\frac{2^E + M}{\eta + \mathbf{e}_{\sigma'(\delta_1)}}, \frac{2^E + M}{-\mathbf{e}_{\min} - \mathbf{e}_{\sigma'(\delta_1)}}\right) \right\rceil_{\mathbb{Z}}$, there exists an L -layer σ network f starting with σ such that*

$$f : \delta_1 \mapsto 0 \text{ and } \mathcal{D}_{f, \delta_1}(g_0) = g_1.$$

Proof. By Condition 2, we have $|w \otimes \sigma(\delta_0)|, |w \otimes \sigma(\delta_1)| < 2^{\mathbf{e}_{\max} + 1}$ for $w \in \mathbb{F}$ with $|w| < 2^{\eta + 1}$. Hence for all $w \in \mathbb{F}$ with $|w| < 2^{\eta + 1}$, we have $AS_{(\delta_1), (\delta_1), w} \neq \emptyset$ by Lemma A.7.

Note that $\sigma'(\delta_1) = \mathbf{s}_{\sigma'(\delta_1)} \times 2^{\mathbf{e}_{\sigma'(\delta_1)}}$ for $\mathbf{s}_{\sigma'(\delta_1)} \in \{1, -1\}$ by Condition 2. We only consider the case $\mathbf{s}_{\sigma'(\delta_1)} = 1$, as the negative case is symmetric.

First, consider the case $\mathbf{m}_{g_0} = \mathbf{m}_{g_1}$. For each $i \in [\mathbf{e}_{\min}, \eta]_{\mathbb{Z}}$, define $w_i := 2^i$ and choose $h_{w_i} \in AS_{(\delta_1), (0), w_i}$. Then $\mathcal{D}_{h_i, \delta_1}(g_0)$ can be calculated as

$$\mathcal{D}_{h_i, \delta_1}(g_0) = g_0 \otimes w_i \otimes \sigma'(\delta_1) = g_0 \otimes 2^i \otimes 2^{\mathbf{e}_{\sigma'(\delta_1)}}.$$

Therefore, we can increase and decrease the exponent by at most $\eta + \mathbf{e}_{\sigma'(\delta_1)}$ and $-\mathbf{e}_{\min} - \mathbf{e}_{\sigma'(\delta_1)}$, respectively. Note that $\eta + \mathbf{e}_{\sigma'(\delta_1)} \geq 1$ by Condition 2.

For any $g_0 = \mathbf{s}_{g_0} \times \mathbf{m}_{g_1} \times 2^{\mathbf{e}_{g_0}}$, there exists $i_1, \dots, i_m \in [\mathbf{e}_{\min}, \eta]_{\mathbb{Z}}$ such that $\sum_{j=1}^m (i_j + \mathbf{e}_{\sigma'(\delta_1)}) = \mathbf{e}_{g_1} - \mathbf{e}_{g_0}$ with $m \geq \max\left(\frac{2^E + M}{\eta + \mathbf{e}_{\sigma'(\delta_1)}}, \frac{2^E + M}{-\mathbf{e}_{\min} - \mathbf{e}_{\sigma'(\delta_1)}}\right)$.

Let $w_j = \begin{cases} 2^{i_j} & \text{if } j \in [m-1] \\ \mathbf{s}_{g_0} \times 2^{i_m} & \text{if } j = m \end{cases}$ and $h_j \in AS_{(\delta_1), (0), w_j}$ and $h := h_1 \circ h_2 \circ \dots \circ h_m$. Then, we have

$$h(\delta_0) = \delta_1, \quad \mathcal{D}_{h, \delta_1}(g_0) = g_1.$$

Now consider the general case ($\mathbf{m}_{g_0} \neq \mathbf{m}_{g_1}$). We now adjust the mantissa by modifying some of the weights 2^{i_j} to $\mathbf{m} \times 2^{i_j}$ to adjust the mantissa. Note that we may target either $2g_1$ or $g_1/2$ as the original maximal exponent gap is $2^E + M - 1$.

By Lemma B.7, there exists $\mathbf{m}_{g_0}^\dagger \in \mathbb{F}$ such that $\mathbf{m}_{g_0}^\dagger \otimes \mathbf{m}_{g_0} = 1^+$.

If $\mathbf{m}_{g_0} = 1$, let $w_j = \begin{cases} 2^{i_j} & \text{if } j \in [m-1] \\ \mathbf{s}_{g_0} \times \mathbf{m}_{g_1} \times 2^{i_m} & \text{if } j = m \end{cases}$.

$$\text{If } \mathfrak{m}_{g_0} = 1^+, \text{ let } w_j = \begin{cases} 2^{i_j} & \text{if } j \in [m-2] \\ 2^- \times 2^{i_{m-1}} & \text{if } j = m-1 \\ \mathfrak{s}_{g_0} \times \mathfrak{m}_{g_1} \times 2^{i_m} & \text{if } j = m \end{cases}.$$

$$\text{If } \mathfrak{m}_{g_0} \neq 1, 1^+, \text{ let } w_j = \begin{cases} 2^{i_j} & \text{if } j \in [m-1] \\ \mathfrak{s}_{g_0} \times \mathfrak{m}_{g_0}^\dagger \times 2^{i_m} & \text{if } j = m \end{cases}.$$

$$\text{If } \mathfrak{m}_{\mathcal{D}_{h_1, \delta_1}(g_0)} = 1^+, \text{ by taking } w_j = \begin{cases} \mathfrak{m}_{g_1} \times 2^{i_j} & \text{if } j = 1 \\ 2^- \times 2^{i_j} & \text{if } j = 2 \\ 2^{i_j} & \text{if } j \in [m-1] \setminus [2] \\ \mathfrak{s}_{g_0} \times \mathfrak{m}_{g_0}^\dagger \times 2^{i_m} & \text{if } j = m \end{cases}, \text{ we have } \mathcal{D}_{h_1, \delta_1}(g_0) = g_1.$$

If $\mathfrak{m}_{\mathcal{D}_{h_1, \delta_1}(g_0)} \neq 1^+$, then $\mathcal{D}_{h_1, \delta_1}(g_0)$ is subnormal and there are two possible cases.

$$\text{If } \mathfrak{m}_{\mathcal{D}_{h_1, \delta_1}(g_0)} = 1, \text{ by taking } w_j = \begin{cases} \mathfrak{m}_{g_1} \times 2^{i_j} & \text{if } j = 1 \\ 2^{i_j} & \text{if } j \in [m-1] \setminus [1] \\ \mathfrak{s}_{g_0} \times \mathfrak{m}_{g_0}^\dagger \times 2^{i_m} & \text{if } j = m \end{cases}, \text{ we have } \mathcal{D}_{h_1, \delta_1}(g_0) = g_1.$$

$$\text{If } \mathfrak{m}_{\mathcal{D}_{h_1, \delta_1}(g_0)} = 1^{++}, \text{ by taking } w_j = \begin{cases} \mathfrak{m}_{g_1} \times 2^{i_j} & \text{if } j = 1 \\ (2 - 2^{-M+2}) \times 2^{i_j} & \text{if } j = 2, M \geq 3 \\ 1^+ \times 2^{i_j} & \text{if } j = 2, M = 2 \\ 2^{i_j} & \text{if } j \in [m-1] \setminus [2] \\ \mathfrak{s}_{g_0} \times \mathfrak{m}_{g_0}^\dagger \times 2^{i_m} & \text{if } j = m \end{cases}, \text{ we have } \mathcal{D}_{h_1, \delta_1}(g_0) = g_1.$$

This completes the proof. □

Proof of Lemma 4.4. Consider $L \geq \left\lceil \max \left(\frac{2^E + M}{\eta + \epsilon_{\sigma'(\delta_1)}}, \frac{2^E + M}{-\epsilon_{\min} - \epsilon_{\sigma'(\delta_1)}} \right) \right\rceil_{\mathbb{Z}}$. For any $k \in [n]$ and $y \in \mathbb{F}$ such that $0 < |y| \leq \Omega \times 2^{\epsilon_{\sigma'(\delta_1)}}$, by Lemma B.8, there exists an L -layer σ network $f_k : \mathbb{F} \rightarrow \mathbb{F}$ starting with σ such that

$$f_k : \delta_1 \mapsto 0 \text{ and } \mathcal{D}_{f, \delta_1}(y) = y_k^*.$$

Then $\tilde{f}_k : x \mapsto f_k(x_k)$ is an L -layer neural network starting with σ . Define f as

$$f := \#_{k=1}^n \tilde{f}_k.$$

Then $f(\delta_1 \otimes \mathbf{1}_n) = 0$ and $\mathcal{D}_{f, \delta_1 \otimes \mathbf{1}_n} = y^*$. This completes the proof. □

B.7. Proof of Lemma 4.5

Lemma B.9. *Let $x, x_0 \in \mathbb{F}$ satisfy*

$$\frac{1}{2} \times 2^{\epsilon_x - M} < x_0 < \frac{3}{2} \times 2^{\epsilon_x - M}.$$

Then there exists a natural number $n \in \mathbb{N}$ such that

$$\bigoplus_{i=1}^n x_0 = |x|.$$

Proof. Consider the largest integer n such that

$$\bigoplus_{i=1}^n x_0 < |x|.$$

Such an n exists, since for any $y \in \mathbb{F}$ such that $y < 2^{\epsilon_x}$,

$$y \oplus x_0 > y.$$

By maximality of n , we have

$$\bigoplus_{i=1}^{n+1} x_0 \geq |x|.$$

On the other hand,

$$\bigoplus_{i=1}^{n+1} x_0 \leq \bigoplus_{i=1}^n x_0 \oplus x_0 \leq |x|^- \oplus x_0 \leq |x|,$$

Therefore,

$$\bigoplus_{i=1}^{n+1} x_0 = x.$$

This completes the proof. $\square$

Lemma B.10. *Consider $x, y \in \mathbb{F}$ satisfying $xy > 0$, $|y| \geq |x|$, and $\mathbf{m}_{y,M} = 0$. Then there exists a floating-point number $b \in \mathbb{F}$ such that*

$$b \oplus x = y,$$

and

$$|b| \leq |y|.$$

Proof. Without loss of generality, assume that $y > x > 0$.

If $\epsilon_x = \epsilon_y$, define

$$b := (\mathbf{m}_y - \mathbf{m}_x) \times 2^{\epsilon_x}.$$

Then $b \oplus x = y$.

Hence, it suffices to consider the case $\epsilon_y > \epsilon_x \geq \epsilon_{\min}$. Define $b_0 \in \mathbb{F}$ as the largest floating-point number such that

$$b_0 \oplus x \leq y.$$

By definition,

$$b_0^+ \oplus x > y.$$

Assume, for the sake of contradiction, that

$$b_0 \oplus x < y.$$

Case 1: $\mathbf{m}_{y,M} = 0$ and $y \neq 2^{\epsilon_y}$. As $b_0 \oplus x < y$, we have

$$b_0 + x < y - 2^{\epsilon_y - M - 1}.$$

Moreover, $b_0^+ \oplus x > y$ implies

$$b_0 + 2^{\epsilon_{b_0} - M} + x > y + 2^{\epsilon_y - M - 1} = (y - 2^{\epsilon_y - M - 1}) + 2^{\epsilon_y - M} > b_0 + x + 2^{\epsilon_y - M}.$$

Therefore,

$$\epsilon_{b_0} > \epsilon_y,$$

which contradicts $b_0 + x < y$.

Case 2: $y = 2^{\epsilon_y}$. From $b_0 \oplus x < y$, we obtain

$$b_0 + x < y - 2^{\epsilon_y - M - 2}.$$

From $b_0^+ \oplus x > y$, we obtain

$$b_0 + 2^{\epsilon_{b_0} - M} + x > y + 2^{\epsilon_y - M - 1}.$$

Combining the two inequalities yields

$$2^{\epsilon_{b_0} - M} > \frac{3}{4} \cdot 2^{\epsilon_y - M}.$$

Hence,

$$\mathfrak{e}_{b_0} = \mathfrak{e}_y.$$

Since $b_0 \leq y$, it follows that

$$b_0 = y = 2^{\mathfrak{e}_y}.$$

Consequently,

$$x < -2^{\mathfrak{e}_y - M - 2},$$

which contradicts $x > 0$. Thus, the assumption $b_0 \oplus x < y$ leads to a contradiction. Therefore,

$$b_0 \oplus x = y.$$

This completes the proof. $\square$

Lemma B.11. Suppose σ satisfies Condition 3 and $\mathcal{X} \subset [-\zeta, \zeta]_{\mathbb{F}}$ is σ -distinguishable with range $[-2^{\mathfrak{e}_{\max}}, 2^{\mathfrak{e}_{\max}}]$. Then, for any $x_0 \in \mathcal{X}$ and $g \in \mathbb{F}$, there exist $n \in \mathbb{N}$, $n_i \in \mathbb{N}$ for $i \in [n]$, and parameters $g_{i,j}, w_i, b_i, w'_{i,j} \in \mathbb{F}$ for $i \in [n]$ and $j \in [n_i]$ such that $|g_{i,j}| \leq 2^{\mathfrak{e}_{\max} + \mathfrak{e}_{\sigma'(\delta_1)}}$, $g = \bigoplus_{i=1}^n \left(\bigoplus_{j=1}^{n_i} g_{i,j} \otimes w'_{i,j} \right) \otimes \sigma'(w_i \otimes x_0 \oplus b_i) \otimes w_i$, and $|w'_{i,j} \otimes \sigma(w_i \otimes x \oplus b_i)| \leq 2^{\mathfrak{e}_{\max}}$, for any $x \in \mathcal{X}$.

Proof. First, for each $i \in [n]$, by Lemma B.9, we can choose appropriate $n_i, w'_{i,j} = 1$, and $g_{i,j} \in \mathbb{F}$ such that $\bigoplus_{j=1}^{n_i} g_{i,j} \otimes w'_{i,j}$ realizes an arbitrary floating-point number. Thus, the problem reduces to find $g'_i, w_i, b_i \in \mathbb{F}$ such that

$$g = \bigoplus_{i=1}^n g'_i \otimes \sigma'(w_i \otimes x_0 \oplus b_i) \otimes w_i,$$

and

$$|\sigma(w_i \otimes x \oplus b_i)| \leq 2^{\mathfrak{e}_{\max}} \text{ for } x \in \mathcal{X}.$$

By Condition 3, there exists $x_1 \in \mathbb{F}$ such that x_1 is even and $|x_1| > \zeta \geq |x_0|$,

$$\sigma([- (\zeta \oplus |x_1|), \zeta \oplus |x_1|]_{\mathbb{F}}) \subset [-2^{\mathfrak{e}_{\max}}, 2^{\mathfrak{e}_{\max}}], \text{ and } |\sigma'(x_1)| \geq 2^{-M},$$

or

$$\sigma([-2\zeta, 2\zeta]_{\mathbb{F}}) \subset [-2^{\mathfrak{e}_{\max}}, 2^{\mathfrak{e}_{\max}}], \text{ and } |\sigma'(0)| \geq 2^{-M}.$$

In the first case, by Lemma B.10, there exists $b \in \mathbb{F}$ such that $b \oplus |x_0| = |x_1|$, and $|b| \leq |x_1|$. For any $|w| \leq 1$, we have $|b \oplus w \otimes x| \leq |x_0| + |x_1| \leq \zeta + |x_1|$ and $|\sigma(b \oplus w \otimes x)| \leq 2^{\mathfrak{e}_{\max}}$ for $x \in \mathcal{X}$.

In the second case, we define x_1 and b as $x_1 := 0$ and $b := -x_0$, respectively. Then $|b \oplus w \otimes x| \leq |x_0| + |x| \leq 2\zeta$ and $|\sigma(b \oplus w \otimes x)| \leq 2^{\mathfrak{e}_{\max}}$ for $x \in \mathcal{X}$.

In both cases, we have $|\sigma(b \oplus w \otimes x)| \leq 2^{\mathfrak{e}_{\max}}$ for $x \in \mathcal{X}$.

Now choose $j_1, j_2 \in [\mathfrak{e}_{\min}, \mathfrak{e}_{\max}]_{\mathbb{Z}}$ such that $j_1 + j_2 = \mathfrak{e}_g - \mathfrak{e}_{\sigma'(x_1)} - M$, $\mathfrak{e}_{\min} - \mathfrak{e}_{\sigma'(x_1)} \leq j_1 \leq \mathfrak{e}_{\max} - \mathfrak{e}_{\sigma'(x_1)}$, and $\mathfrak{e}_{\min} + 1 \leq j_2 \leq 0$.

By Lemma B.7, there exists $\mathfrak{m} \in \mathbb{F}$ such that $\mathfrak{m} \otimes \mathfrak{m}_{\sigma'(x_1)} = 1^+$.

Define g'_i and w_i as $g'_i := \mathfrak{s}_g \times \mathfrak{m} \times 2^{j_1}$ and $w_i := 2^{j_2} \leq 1$. By Lemma B.10, there exist $b_i \in \mathbb{F}$ such that $b_i \oplus (w_i \otimes x_0) = x_1$. Then

$$g'_i \otimes \sigma'(w_i \otimes x_0 \oplus b_i) \otimes w_i = g'_i \otimes \sigma'(x_1) \otimes w_i = \mathfrak{s}_g \times (1^+ \times 2^{j_1 + \mathfrak{e}_{\sigma'(x_1)}}) \otimes 2^{j_2} = \mathfrak{s}_g \times [1^+ \times 2^{\mathfrak{e}_g - M}].$$

By Lemma B.9, there exists a natural number n such that

$$\bigoplus_{i=1}^n g'_i \otimes \sigma'(w_i \otimes x_0 \oplus b_i) \otimes w_i = \bigoplus_{i=1}^n g'_i \otimes \sigma'(x_1) \otimes w_i = g.$$

This completes the proof. $\square$

Proof of Lemma 4.5. By Lemma B.11, for any $x_0 = (x_{0,1}, \dots, x_{0,d}) \in \mathcal{X}$, there exists $g_{i,j}, w_i, b_i, w'_{i,j}$ such that $(y^*)_k = \bigoplus_{i=1}^n \left(\bigoplus_{j=1}^{n_i} g_{i,j} \otimes w'_{i,j} \right) \otimes \sigma'(w_i \otimes x_{0,k} \oplus b_i) \otimes w_i$ with $|g_{i,j}| \leq 2^{\epsilon_{\max} + \epsilon_{\sigma'(\delta_1)}}$. Observe that the right-hand side coincides with the output gradient of a network $H_k : \mathbb{F}^d \rightarrow \mathbb{F}^{\sum_{i=1}^n n_i}$ whose output is the concatenation of n blocks, where the i -th block is the n_i -dimensional vector computed by

$$h_i : x \mapsto \left\| \bigoplus_{j=1}^{n_i} (w_{i,j} \otimes \sigma(w_j \otimes x_k \oplus b_j)) \right\| \in \mathbb{F}^{n_i},$$

where $x = (x_1, \dots, x_d)$. Consider a sequential addition $s : (-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}) \xrightarrow{\sigma} (\delta_1, \delta_1)$. We define H_k by

$$H_k := \left\| \bigoplus_{i=1}^n s \circ h_i \right\|.$$

Then we can check that

$$H_k : \mathcal{X} \mapsto (\delta_1, \dots, \delta_1).$$

Note that when the input gradient of H_k corresponding to the (i, j) -th output coordinate is $g_{i,j}$ and $x_k = x_{0,k}$, the resulting output gradient is $(y^*)_k \otimes e_k$. As H_k depends only on the k -th input coordinate, the function $\tilde{H}$ defined by

$$\tilde{H} := \left\| \bigoplus_{k=1}^d H_k \right\|, \quad (9)$$

is the desired function. □

B.8. Proof of Lemma 4.6

Proof. We first assume $c_1 \leq c_2$. In other case, we can apply similar argument with negative weight.

Consider $\phi_1 \in AS_{(\delta_0, \delta_1), (\delta_0, \delta_1), 2^{\epsilon_{\sigma'(\delta_1)}}}$ and $\phi_2 \in AS_{(\delta_0, \delta_1), (\delta_0, \delta_0), 2^{\epsilon_{\sigma'(\delta_1)}}}$. Then we have

$$\begin{aligned} \phi_1(\delta_0) &= \delta_0, & \phi_2(\delta_0) &= \delta_0, \\ \phi_1(\delta_1) &= \delta_1, & \phi_2(\delta_1) &= \delta_0. \end{aligned}$$

By Condition 2, $\sigma(\delta_0) \neq \sigma(\delta_1)$ and $2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\delta_0), 2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\delta_0) \in \mathbb{F}$, hence

$$2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\delta_1) \ominus 2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\delta_0) \neq 0.$$

Then consider a sequential addition s such that

$$s : (0, 2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\delta_1) \ominus 2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\delta_0)) \xrightarrow{\sigma} (c_1, c_2).$$

Now define the function f as

$$f(x) = s \left(2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\phi_1(x)) \ominus 2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \sigma(\phi_2(x)) \right),$$

and we have

$$f(\delta_0) = s(0) = c_1, \quad f(\delta_1) = s(2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \delta_1 \ominus 2^{-\epsilon_{\sigma'(\delta_1)}} \otimes \delta_0) = c_2.$$

For any $y \in [-\Omega \times 2^{\epsilon_{\sigma'(\delta_1)}}, \Omega \times 2^{\epsilon_{\sigma'(\delta_1)}}]_{\mathbb{F}}$, we have

$$\begin{aligned} \mathcal{D}_{f, \delta_0}(y) &= 0, \\ \mathcal{D}_{f, \delta_1}(y) &= (y \otimes \sigma'(\delta_1) \otimes \sigma'(\delta_1)) \ominus (y \otimes \sigma'(\delta_1) \otimes \sigma'(\delta_0)), \\ 0 &< |\mathcal{D}_{f, \delta_1}(y)| \leq y. \end{aligned}$$

This completes the proof. □

Proof. Note that $\sigma'(\delta_1) = \mathfrak{s}_{\sigma'(\delta_1)} \times 2^{\epsilon_{\sigma'(\delta_1)}}$ for $\mathfrak{s}_{\sigma'(\delta_1)} \in \{1, -1\}$ by Condition 2. Consider $w_1 = \mathfrak{s}_{\sigma'(\delta_1)} \times 2^{-\epsilon_{\delta_1}}$ and let $f_1 \in AS_{(\delta_0, \delta_1), (\delta_0, \delta_1), w_1}$. Then $\mathcal{D}_{f_1, x}(g)$ is given by

$$\mathcal{D}_{f_1, x}(g) = \begin{cases} g \otimes (\mathfrak{s}_{\sigma'(\delta_1)} \times 2^{-\epsilon_{\sigma'(\delta_1)}}) \otimes (\mathfrak{s}_{\sigma'(\delta_1)} \times 2^{\epsilon_{\sigma'(\delta_1)}}) = g & \text{if } x = \delta_0, \\ g \otimes (\mathfrak{s}_{\sigma'(\delta_1)} \times 2^{-\epsilon_{\sigma'(\delta_1)}}) \otimes \sigma'(\delta_0) & \text{if } x = \delta_1. \end{cases}$$

Moreover, we have

$$|g \otimes (\mathfrak{s}_{\sigma'(\delta_1)} \times 2^{-\epsilon_{\sigma'(\delta_1)}}) \otimes \sigma'(\delta_0)| = |(g \times 2^{-\epsilon_{\sigma'(\delta_1)}}) \otimes \sigma'(\delta_0)| \leq g \otimes 2^{-e_0}.$$

by Condition 2. Therefore, for $n \in \mathbb{N}$ satisfying $n \geq (2^E + M)/e_0$, we obtain

$$\mathcal{D}_{f_1^{(n)}, x}(g) = \begin{cases} 0 & \text{if } x = \delta_0, \\ g & \text{if } x = \delta_1. \end{cases}$$

Finally, note that the output values can be replaced by any $c_1, c_2 \in \mathbb{F}$ satisfying $|c_1|, |c_2|, |c_1 - c_2| \leq 2^{\epsilon_{\max}}$ by composing a sequential addition in the final layer. This completes the proof. $\square$

C. Proofs for Real Activation Functions

Lemma C.1. For $a, b \in \mathbb{R}$ with $2^{\epsilon_{\min}} \leq a \leq 2^{-k}b \leq b \leq 2^{\epsilon_{\max}}$ for some $k \in \mathbb{N}_{\geq 0}$, then $\lceil a \rceil \leq 2^k \lceil b \rceil$.

Proof. Since $2^{\epsilon_{\min}} \leq b$, $2^{-k}b \leq 2^{\epsilon_{\max}}$, we have $2^{-k} \lceil b \rceil = \lceil 2^{-k}b \rceil$ and we get the desired result. $\square$

C.1. Proof of Theorem 3.1

To prove Theorem 3.1, we use Lemmas C.2–C.4 presented below.

First note that, by Lemmas C.2–C.4, $\mathcal{X} = [-M_\sigma, M_\sigma]_{\mathbb{F}}$ is $\lceil \hat{\sigma} \rceil$ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{F}}$ for $\sigma \in \Sigma$.

Suppose $\sigma \in \{\lceil \text{ReLU} \rceil, \lceil \text{ELU} \rceil, \lceil \text{GELU} \rceil, \lceil \text{Swish} \rceil\}$. By Lemma C.2, σ satisfies Conditions 1–3 with $\kappa = \epsilon_{\max} + 1$, $\delta_0 = -2^{E+1}$, $\sigma'(\delta_1) = 1$, $\eta = \epsilon_{\max} - 1$, $\zeta = 2^{\epsilon_{\max}-2}$. Note that τ (Eq. (4) in Section 3.3) is

$$\begin{aligned} \tau &:= \left\lceil \max \left(\frac{2^E + M}{\eta + \epsilon_{\sigma'(\delta_1)}} + 4, \frac{2^E + M}{-\epsilon_{\min} - \epsilon_{\sigma'(\delta_1)}} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = \left\lceil \max \left(\frac{2^E + M}{\epsilon_{\max} - 1} + 4, \frac{2^E + M}{-\epsilon_{\min}} + 4, 7 \right) \right\rceil_{\mathbb{Z}} \\ &= \left\lceil \max \left(\frac{2^E + M}{2^{E-1} - 2} + 4, \frac{2^E + M}{2^{E-1} - 2} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = 7, \\ \min \left(\Omega \times 2^{\sigma'(\delta_1)}, 2^\kappa \right) &= \min \left(\Omega, 2^{\epsilon_{\max}+1} \right) = \Omega, \end{aligned}$$

since $2^E + M \leq 5 \cdot 2^{E-2} \leq 3 \cdot (2^{E-1} - 2)$ for $E \geq 6$.

By Theorem 3.6, for any $L \geq 7$, there exists an L -layer σ network f such that $f(x) = f^*(x)$ and $\mathcal{D}_{f, x}(h^*(x)) = g^*(x)$ for $|h^*(x)| \leq \Omega = H_\sigma$ and all $x \in \mathcal{X}$.

Suppose σ is $\lceil \text{Sigmoid} \rceil$. By Lemma C.3, σ satisfies Conditions 1–3 with $\kappa = M - 1$, $\lceil \hat{\sigma}' \rceil(\delta_1) = \frac{1}{4}$, $\eta = \epsilon_{\max}$, $\zeta = 2^{\epsilon_{\max}-1}$. Then we have

$$\begin{aligned} \tau &:= \left\lceil \max \left(\frac{2^E + M}{\eta + \epsilon_{\sigma'(\delta_1)}} + 4, \frac{2^E + M}{-\epsilon_{\min} - \epsilon_{\sigma'(\delta_1)}} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = \left\lceil \max \left(\frac{2^E + M}{\epsilon_{\max} - 3} + 4, \frac{2^E + M}{-\epsilon_{\min} - 2} + 4, 7 \right) \right\rceil_{\mathbb{Z}} \\ &= \left\lceil \max \left(\frac{2^E + M}{2^{E-1} - 4} + 4, \frac{2^E + M}{2^{E-1} - 4} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = 7, \\ \min \left(\Omega \times 2^{\sigma'(\delta_1)}, 2^\kappa \right) &= \min \left(\Omega \times 2^{-2}, 2^{M-1} \right) = 2^{M-1}, \end{aligned}$$

since $2^E + M \leq 5 \cdot 2^{E-2} \leq 3 \cdot (2^{E-1} - 4)$ for $E \geq 6$.

By Theorem 3.6, for any $L \geq 7$, there exists an L -layer σ network f such that $f(x) = f^*(x)$ and $\mathcal{D}_{f,x}(h^*(x)) = g^*(x)$ for $|h^*(x)| \leq 2^{M-1}$ all $x \in \mathcal{X}$.

Suppose σ is $\lceil \tanh \rceil$. By Lemma C.4 σ satisfies Conditions 1–3 with $\kappa = M - 2$, $\lceil \hat{\sigma}' \rceil(\delta_1) = 1$, $\eta = \epsilon_{\max} - 1$. Then we have

$$\begin{aligned} \tau &:= \left\lceil \max \left(\frac{2^E + M}{\eta + \epsilon_{\sigma'(\delta_1)}} + 4, \frac{2^E + M}{-\epsilon_{\min} - \epsilon_{\sigma'(\delta_1)}} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = \left\lceil \max \left(\frac{2^E + M}{\epsilon_{\max} - 1} + 4, \frac{2^E + M}{-\epsilon_{\min}} + 4, 7 \right) \right\rceil_{\mathbb{Z}} \\ &= \left\lceil \max \left(\frac{2^E + M}{2^{E-1} - 2} + 4, \frac{2^E + M}{2^{E-1} - 2} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = 7, \\ \min \left(\Omega \times 2^{\sigma'(\delta_1)}, 2^\kappa \right) &= \min \left(\Omega, 2^{M-2} \right) = 2^{M-2}, \end{aligned}$$

since $2^E + M \leq 5 \cdot 2^{E-2} \leq 3 \cdot (2^{E-1} - 2)$ for $E \geq 6$.

By Theorem 3.6, for any $L \geq 7$, there exists an L -layer σ network f such that $f(x) = f^*(x)$ and $\mathcal{D}_{f,x}(h^*(x)) = g^*(x)$ for all $x \in \mathcal{X}$.

Lemma C.2. *Let $\hat{\sigma}$ be one of ReLU, ELU, GELU, Swish, ELU. Then $\mathcal{X} = [-2^{\epsilon_{\max}-2}, 2^{\epsilon_{\max}-2}]_{\mathbb{R}}$ is $\lceil \hat{\sigma} \rceil$ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{R}}$ and $\lceil \hat{\sigma} \rceil$ satisfies Conditions 1–3 with $\kappa = \epsilon_{\max} + 1$, $\delta_0 = -2^E$, $\lceil \hat{\sigma}' \rceil(\delta_1) = 1$, $\eta = \epsilon_{\max} - 1$, $\zeta = 2^{\epsilon_{\max}-2}$.*

Lemma C.3. *Let $\hat{\sigma}$ be Sigmoid. Then $\mathcal{X} = [-2^{\epsilon_{\max}-1}, 2^{\epsilon_{\max}-1}]_{\mathbb{R}}$ is $\lceil \hat{\sigma} \rceil$ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{R}}$ and $\lceil \hat{\sigma} \rceil$ satisfies Conditions 1–3 with $\kappa = M - 1$, $\lceil \hat{\sigma}' \rceil(\delta_1) = \frac{1}{4}$, $\eta = \epsilon_{\max} - 1$, $\zeta = 2^{\epsilon_{\max}-1}$.*

Lemma C.4. *Let $\hat{\sigma}$ be tanh. Then $\mathcal{X} = [-2^{\epsilon_{\max}-1}, 2^{\epsilon_{\max}-1}]_{\mathbb{R}}$ is $\lceil \hat{\sigma} \rceil$ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{R}}$ and $\lceil \hat{\sigma} \rceil$ satisfies Conditions 1–3 with $\kappa = M - 2$, $\lceil \hat{\sigma}' \rceil(\delta_1) = 1$, $\eta = \epsilon_{\max} - 1$, $\zeta = 2^{\epsilon_{\max}-1}$.*

The detailed proofs of Lemmas C.2–C.4 are presented in Appendices C.5–C.7.

C.2. Proof of Theorem 3.2

From the proof of Theorem 3.1, we observe that σ satisfies Conditions 1–3 with $\tau = 7$. Hence, by Theorem 3.8, for $L \geq 2^{E+1} + 2M + 9$, there exists an L -layer σ network f such that $f(x) = f^*(x)$ and $\mathcal{D}_{f,x}(y) = g^*(x, y)$ for all $x \in \mathcal{X}$ and $y \in [-H_\sigma, H_\sigma]_{\mathbb{R}}$.

C.3. The case of $4 \leq E \leq 5$ in Theorems 3.1 and 3.2

In the case $E = 5$, the number of layers required in Theorem 3.1 increases to 8 when $\sigma = \lceil \text{Sigmoid} \rceil$, while it remains unchanged for the other activations. In the case $E = 4$, the number of layers required in Theorem 3.1 increases by 1 for all activations in Σ .

For Theorem 3.2 with $E = 5$, the required number of layers increases to $2^{E+1} + 2M + 10$ when $\sigma = \lceil \text{Sigmoid} \rceil$, while it remains unchanged for the other activations. In the case $E = 4$, the number of layers required in Theorem 3.2 increases by 1 for all activations in Σ .

Suppose $\sigma \in \{\lceil \text{ReLU} \rceil, \lceil \text{ELU} \rceil, \lceil \text{GELU} \rceil, \lceil \text{Swish} \rceil\}$. Then τ is

$$\tau = \left\lceil \max \left(\frac{2^E + M}{2^{E-1} - 2} + 4, \frac{2^E + M}{2^{E-1} - 2} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = \begin{cases} 8 & \text{if } E = 4 \\ 7 & \text{if } E = 5 \end{cases},$$

Suppose σ is $\lceil \text{Sigmoid} \rceil$. Then we have

$$\tau = \left\lceil \max \left(\frac{2^E + M}{2^{E-1} - 4} + 4, \frac{2^E + M}{2^{E-1} - 4} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = \begin{cases} 9 & \text{if } E = 4 \\ 8 & \text{if } E = 5 \end{cases},$$

Suppose σ is $\lceil \tanh \rceil$. Then we have

$$\tau = \left\lceil \max \left(\frac{2^E + M}{2^{E-1} - 2} + 4, \frac{2^E + M}{2^{E-1} - 2} + 4, 7 \right) \right\rceil_{\mathbb{Z}} = \begin{cases} 8 & \text{if } E = 4 \\ 7 & \text{if } E = 5 \end{cases},$$

Corollary C.5. Let $4 \leq E \leq 5$, $\sigma \in \Sigma$, $\mathcal{X} = [-M_\sigma, M_\sigma]_{\mathbb{F}}^d$, $f^* : \mathcal{X} \rightarrow \mathbb{F}$, $h^* : \mathcal{X} \rightarrow [-H_\sigma, H_\sigma]_{\mathbb{F}}$, and $g^* : \mathbb{F}^d \rightarrow \mathbb{F}^d$ such that $g^*(x) = 0$ for all $x \in \mathcal{X}$ with $h^*(x) = 0$. Then, for any $L \geq \tau$, there exists an L -layer σ network f such that $f(x) = f^*(x)$ and $\mathcal{D}_{f,x}(h^*(x)) = g^*(x)$ for all $x \in \mathcal{X}$ where

$$\tau = \begin{cases} 13 & \text{if } E = 4, \sigma = [\text{Sigmoid}], \\ 10 & \text{if } E = 4, \sigma \in \Sigma_\sigma \setminus \{[\text{Sigmoid}]\} \text{ or } E = 5, \sigma \in \{\Sigma \setminus \{[\tanh]\}\} \\ 9 & \text{if } E = 5, \sigma = \tanh. \end{cases}$$

Corollary C.6. Let $4 \leq E \leq 5$, $\sigma \in \Sigma$, $\mathcal{X} = [-M_\sigma, M_\sigma]_{\mathbb{F}}^d$, $f^* : \mathcal{X} \rightarrow \mathbb{F}$, and $g^* : \mathbb{F}^d \times \mathbb{F} \rightarrow \mathbb{F}^d$ such that $g^*(x, -y) = -g^*(x, y)$. Then, for any $L \geq 2^{E+1} + 2M + 2 + \tau$, there exists an L -layer σ network f such that $f(x) = f^*(x)$ and $\mathcal{D}_{f,x}(y) = g^*(x, y)$ for all $x \in \mathcal{X}$ and $y \in [-H_\sigma, H_\sigma]_{\mathbb{F}}$ where

$$\tau = \begin{cases} 13 & \text{if } E = 4, \sigma = [\text{Sigmoid}], \\ 9 & \text{if } E = 5, \sigma = \tanh. \end{cases}$$

C.4. Lemmas about Conditions 1–3 on real activation functions

Lemma C.7. Suppose a continuous function $\hat{\sigma} : \mathbb{R} \rightarrow \mathbb{R}$ is C^2 on $(0, \infty)$ and satisfies the following with $\frac{1}{2} \leq c_1 \leq 1 \leq c_2 \leq 2$:

(A1) $\hat{\sigma}(x)$ is increasing and $c_1 x \leq \hat{\sigma}(x) \leq c_2 x$ for $x \geq 0$.

(A2) $\frac{1}{2} \leq \hat{\sigma}'(x) \leq \frac{3}{2}$ for $x \geq 0$.

(A3) $\begin{cases} 1 \in \hat{\sigma}'([\frac{1}{2}, 1]) \text{ and } |\hat{\sigma}''(x)| \leq \frac{3}{4} \text{ for } \frac{1}{2} \leq x \leq 1 \\ 1 \in \hat{\sigma}'([1, 2]) \text{ and } |\hat{\sigma}''(x)| \leq \frac{2}{5} \text{ for } 1 \leq x \leq 2. \end{cases}$

(A4) $|\hat{\sigma}(x)| \leq 1$ for $x \leq 0$, and $|\hat{\sigma}'(x)| \leq 2^x$ for $x \leq -10$.

Then $\lceil \hat{\sigma} \rceil = \sigma$ satisfies Condition 1 with $\nu = 5$, $\gamma_1 = 2^{-M-1}$, $\gamma_2 = 2^{-3}$, $\gamma_3 = 2$, $\gamma_4 = 2^4$, $\gamma_5 = (2 - 2^{-M}) \times 2^{\mathfrak{e}_{\max} - M + 2}$, $\kappa = \mathfrak{e}_{\max} + 1$, satisfies Condition 2 with $\delta_0 = -2^{E+1}$, $\eta = \mathfrak{e}_{\max} - 1$ and some $\delta_1 \in (\frac{1}{2}, 2)_{\mathbb{F}}$, and satisfies Condition 3 with $\zeta = 2^{\mathfrak{e}_{\max} - 2}$.

Proof. Condition 1 :

By (A1), we have $\hat{\sigma}(0) = 0$ leading to $\gamma_0 = 0$ and $\lceil \hat{\sigma} \rceil(\gamma_0) = 0$ and $\lceil \hat{\sigma}' \rceil(\gamma_0) = 0$ by (A2).

Let $\nu = 5$, $\gamma_1 = 2^{-M-1}$, $\gamma_2 = 2^{-3}$, $\gamma_3 = 2$, $\gamma_4 = 2^4$, $\gamma_5 = (2 - 2^{-M}) \times 2^{\mathfrak{e}_{\max} - M + 2}$, $\kappa = \mathfrak{e}_{\max} + 1$.

By (A2), we have

$$\begin{aligned} \frac{1}{2} \leq |\hat{\sigma}'(\gamma_i)| &\leq \frac{3}{2} \leq 2^{-k_i} c_1 \gamma_i \leq 2^{-k_i} |\hat{\sigma}(\gamma_i)|, \quad 2^{k_i} \leq \frac{1}{3} 2^{\mathfrak{e}_{\gamma_i}}, \quad k_i \leq \mathfrak{e}_{\gamma_i} - 2, \quad i = 1, \dots, 4 \\ \frac{1}{2} \leq |\hat{\sigma}'(\gamma_5)| &\leq \frac{3}{2} \leq 2^{-k_5} c_1 \gamma_5 \leq 2^{-k_5} |\hat{\sigma}(\gamma_i)|, \quad 2^{k_5} \leq \frac{1}{3} \times (2 - 2^{-M}) \times 2^{\mathfrak{e}_{\max} - M + 2}, \quad k_i \leq \mathfrak{e}_{\max} - M + 1, \end{aligned}$$

By Lemma C.1, we have

$$|\lceil \hat{\sigma}' \rceil(\gamma_i)| \leq 2^{-k_i} |\lceil \hat{\sigma} \rceil(\gamma_i)|.$$

Hence we pick $k_1 = -M - 3$, $k_2 = -5$, $k_3 = -1$, $k_4 = 2$, $k_5 = \mathfrak{e}_{\max} - M + 1$.

Since $\mathfrak{e}_{\min} \leq -M - 1$, $\mathfrak{e}_{\min} \leq -6$, $\mathfrak{e}_{\gamma_i} - 1 \leq \mathfrak{e}_{\hat{\sigma}(\gamma_i)} \leq \mathfrak{e}_{\gamma_i} + 1$ by (A1), we have

$$\begin{aligned} \bigcup_{i=1}^{\nu} [\mathfrak{e}_{\hat{\sigma}(\gamma_i)} + \mathfrak{e}_{\min}, \mathfrak{e}_{\max} - \kappa + \min(\mathfrak{e}_{\hat{\sigma}(\gamma_i)}, k_i)] \supset \\ [\mathfrak{e}_{\min} - M, -M - 4]_{\mathbb{Z}} \cup [\mathfrak{e}_{\min} - 2, -6]_{\mathbb{Z}} \cup [\mathfrak{e}_{\min} + 2, -2]_{\mathbb{Z}} \cup [\mathfrak{e}_{\min} + 5, 1]_{\mathbb{Z}} \cup [-M + 4, \mathfrak{e}_{\max} - M]_{\mathbb{Z}} \end{aligned}$$

$$= [\epsilon_{\min} - M, \epsilon_{\max} - M].$$

Condition 2 :

Suppose there is no $\delta_1 \in [1, 2]_{\mathbb{F}}$ such that $\lceil \hat{\sigma}' \rceil(\delta_1) = 1$. Since $\hat{\sigma}$ satisfies (A3), we consider following cases.

Case 1: $1 \in \hat{\sigma}'([\frac{1}{2}, 1])$ and $|\hat{\sigma}''(x)| \leq \frac{3}{4}$ for $\frac{1}{2} \leq x \leq 1$.

Case 2: $1 \in \hat{\sigma}'([1, 2])$ and $|\hat{\sigma}''(x)| \leq \frac{2}{5}$ for $1 \leq x \leq 2$.

$$\text{Let } [a, b] = \begin{cases} [\frac{1}{2}, 1] & \text{if Case 1,} \\ [1, 2] & \text{if Case 2.} \end{cases}, M_* = \begin{cases} \frac{3}{4} & \text{if Case 1,} \\ \frac{2}{5} & \text{if Case 2.} \end{cases}$$

Let $A := \{x \in [a, b]_{\mathbb{F}} : \lceil \hat{\sigma}' \rceil(y) < 1 \text{ for } y \in [a, x]_{\mathbb{F}}\}$. We claim that $A \neq \emptyset, [a, b]_{\mathbb{F}}$.

First suppose $A = \emptyset$, i.e. $\lceil \hat{\sigma}' \rceil(x) > 1$ for all $x \in [a, b]_{\mathbb{F}}$. By (A3), pick $\xi \in [a, b] \setminus [a, b]_{\mathbb{F}}$ such that $\hat{\sigma}'(\xi) = 1$. Then there exist $\theta \in [a, b]_{\mathbb{F}}$ such that $\theta < \xi < \theta^+$ with $\lceil \hat{\sigma}' \rceil(\theta), \lceil \hat{\sigma}' \rceil(\theta^+) > 1$. Then we have $\hat{\sigma}'(\theta), \hat{\sigma}'(\theta^+) > 1 + 2^{-M-1}$. By mean value theorem, there exist $t_1, t_2 \in (\theta, \theta^+)$ such that

$$\hat{\sigma}'(\xi) - \hat{\sigma}'(\theta) = \hat{\sigma}''(t_1)(\xi - \theta), \quad \hat{\sigma}'(\theta^+) - \hat{\sigma}'(\theta) = \hat{\sigma}''(t_2)(\theta^+ - \xi).$$

Since $(\xi - \theta) + (\theta^+ - \xi) = \theta^+ - \theta \leq \begin{cases} 2^{-M-1} & \text{if Case 1} \\ 2^{-M} & \text{if Case 2} \end{cases}$, we have

$$(\xi - \theta) \leq \begin{cases} 2^{-M-2} & \text{if Case 1,} \\ 2^{-M-1} & \text{if Case 2,} \end{cases} \text{ or } (\theta^+ - \xi) \leq \begin{cases} 2^{-M-2} & \text{if Case 1,} \\ 2^{-M-1} & \text{if Case 2.} \end{cases}$$

Since

$$|\hat{\sigma}''(t_1)| = \frac{\hat{\sigma}'(\xi) - \hat{\sigma}'(\theta)}{\xi - \theta} > \frac{2^{-M-1}}{\xi - \theta}, \quad |\hat{\sigma}''(t_2)| > \frac{2^{-M-1}}{\theta^+ - \xi},$$

we have

$$|\hat{\sigma}''(t_1)| > \begin{cases} 2 & \text{if Case 1,} \\ 1 & \text{if Case 2,} \end{cases} \text{ or } |\hat{\sigma}''(t_2)| > \begin{cases} 2 & \text{if Case 1,} \\ 1 & \text{if Case 2,} \end{cases}$$

which is contradiction to (A3). Hence $A \neq \emptyset$.

Next suppose $A = [a, b]_{\mathbb{F}}$, i.e. $\lceil \hat{\sigma}' \rceil(x) < 1$ for all $x \in [a, b]_{\mathbb{F}}$. Pick $\xi \in [a, b] \setminus [a, b]_{\mathbb{F}}$ such that $\hat{\sigma}'(\xi) = 1$. Then there exist $\theta \in [a, b]_{\mathbb{F}}$ such that $\theta < \xi < \theta^+$ with $\lceil \hat{\sigma}' \rceil(\theta), \lceil \hat{\sigma}' \rceil(\theta^+) < 1$. Then we have $\hat{\sigma}'(\theta), \hat{\sigma}'(\theta^+) < 1 - 2^{-M-2}$.

Since $(\xi - \theta) + (\theta^+ - \xi) = \theta^+ - \theta \leq \begin{cases} 2^{-M-1} & \text{if Case 1} \\ 2^{-M} & \text{if Case 2} \end{cases}$, we have

$$(\xi - \theta) \leq \begin{cases} 2^{-M-2} & \text{if Case 1,} \\ 2^{-M-1} & \text{if Case 2,} \end{cases} \text{ or } (\theta^+ - \xi) \leq \begin{cases} 2^{-M-2} & \text{if Case 1,} \\ 2^{-M-1} & \text{if Case 2.} \end{cases}$$

Since

$$|\hat{\sigma}''(t_1)| = \frac{\hat{\sigma}'(\xi) - \hat{\sigma}'(\theta)}{\xi - \theta} > \frac{2^{-M-2}}{\xi - \theta}, \quad |\hat{\sigma}''(t_2)| > \frac{2^{-M-2}}{\theta^+ - \xi},$$

we have

$$|\hat{\sigma}''(t_1)| > \begin{cases} 1 & \text{if Case 1,} \\ \frac{1}{2} & \text{if Case 2,} \end{cases} \text{ or } |\hat{\sigma}''(t_2)| > \begin{cases} 1 & \text{if Case 1,} \\ \frac{1}{2} & \text{if Case 2,} \end{cases}$$

which is contradiction to (A3).

Therefore $A \neq \emptyset, [a, b]_{\mathbb{F}}$. Let $\xi_* := \max A < b$.

Then by (A3), we have $\lceil \hat{\sigma}' \rceil(\xi_*^+) > 1$ and $\hat{\sigma}'(\xi_*^+) > 1 + 2^{-M-1}$.

$$\hat{\sigma}'(\xi_*) \geq \hat{\sigma}'(\xi_*^+) - \max_{x \in (a, b)} |\hat{\sigma}''(x)|(\xi_*^+ - \xi_*) \geq \hat{\sigma}'(\xi_*^+) - M_*(\xi_*^+ - \xi_*) > 1 - 2^{-M-2},$$

leading to $\lceil \hat{\sigma}' \rceil(\xi_*) \geq 1$ which is a contradiction. Therefore, we have $\delta_1 \in [1, 2]_{\mathbb{F}}$ such that $\lceil \hat{\sigma}' \rceil(\delta_1) = 1$.

By (A1), $\frac{1}{2} \leq \hat{\sigma}(\delta_1) \leq 2$.

Let $\delta_0 = -2^{E+1}$ and $\eta = \epsilon_{\max} - 1$. Since $-2^{2^{E-1}-1} \leq \delta_0 \leq -10$, $\delta_0 \in \mathbb{F}$. By (A4), we have $|\hat{\sigma}'(\delta_0)| \leq 2^{-2^{E+1}} \leq 2^{-8M}$. By (A2) and (A4),

$$\begin{aligned} |\delta_0|, |\delta_1|, |\delta_0 - \delta_1| &\leq 2^{E+1} + 2 \leq 2^{\epsilon_{\max}}, \\ |\lceil \hat{\sigma} \rceil(\delta_0)|, |\lceil \hat{\sigma} \rceil(\delta_1)|, |\lceil \hat{\sigma} \rceil(\delta_0) - \lceil \hat{\sigma} \rceil(\delta_1)| &\leq 5 < 8 \leq 2^{\epsilon_{\max}}, \\ -2^{-M-1} \lceil \hat{\sigma}' \rceil(\delta_1) = -2^{-M-1} &\leq 2 \lceil \hat{\sigma}' \rceil(\delta_0) \leq 1 = \lceil \hat{\sigma}' \rceil(\delta_1), \\ \max(|\lceil \hat{\sigma} \rceil(\delta_0)|, |\lceil \hat{\sigma} \rceil(\delta_1)|) &\leq 2 = 2^{\epsilon_{\max} - \eta}. \end{aligned}$$

Condition 3 :

Take $\zeta = 2^{\epsilon_{\max}-2}$. Now we verify the first item in Condition 3. Let $x_0 \in [-\zeta, \zeta]$ and take $x = \zeta$ and x is even. By (A2), $|\lceil \hat{\sigma}' \rceil(x)| \geq \frac{1}{2} \geq 2^{-M}$. By (A1), we have

$$x \oplus \zeta = 2\zeta \in \mathbb{F}, \quad \zeta \oplus |x| = 2\zeta, \quad \hat{\sigma}([-(\zeta \oplus |x|), 0]) \subset [0, \frac{1}{4}], \quad \hat{\sigma}([0, \zeta \oplus |x|]) \subset [0, 4\zeta] \subset [0, 2^{\epsilon_{\max}}].$$

Therefore,

$$\lceil \hat{\sigma} \rceil([-(\zeta \oplus |x|), (\zeta \oplus |x|)]) \subset [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}].$$

□

Lemma C.8. Suppose continuous function $\hat{\sigma} : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following with $\frac{1}{2} \leq c_1 \leq 1 \leq c_2 \leq 2$:

(B1) $4^x \leq |\hat{\sigma}(x)|, |\hat{\sigma}'(x)| \leq 2^x$ for $x \leq -2$.

(B2) $c_1 \leq |\hat{\sigma}(x)| \leq c_2, \hat{\sigma}'(x) \leq 1$ for $x \geq 0$.

(B3) $\hat{\sigma}(x)$ is increasing and $\hat{\sigma}'(0) \geq \frac{1}{4}$.

(B4) $|\hat{\sigma}''(x)| \leq \frac{1}{10}$ and is C^2 on $(0, \infty)$.

(B5) $\frac{1}{4} \in \hat{\sigma}'([0, 2])$.

(B6) $|\hat{\sigma}(x)| \leq 1$ for $x \leq 0$.

Then $\lceil \hat{\sigma} \rceil = \sigma$ satisfies Condition 1 with $\kappa = M - 1$, satisfies Condition 2 with $\eta = \epsilon_{\max} - 1$, and satisfies Condition 3 with $\zeta = 2^{\epsilon_{\max}-1}$.

Proof. Condition 1 :

Let $\gamma_0 = -2^E$. Since $-2^E \geq -2^{\epsilon_{\max}} = -2^{2^{E-1}-1}$ for $E \geq 3$, $\gamma_0 \in \mathbb{F}$. By (B1), we have

$$|\hat{\sigma}(\gamma_0)|, |\hat{\sigma}'(\gamma_0)| \leq 2^{-2^E} \leq 2^{-2^{E-1}-M+1} = 2^{\epsilon_{\min}-M-1} = \frac{\omega}{2},$$

leading to $\lceil \hat{\sigma} \rceil(\gamma_0) = 0$ and $\lceil \hat{\sigma}' \rceil(\gamma_0) = 0$.

Let $\nu = M - 2, \kappa = M - 1$. Pick $\gamma_1, \gamma_2, \dots, \gamma_{M-3}, \gamma_{M-2} \in \mathbb{F}$ such that $\gamma_1, \dots, \gamma_{M-3} < 0, \gamma_{M-2} > 0$ and

$$2^{-M-2+i} \leq |\hat{\sigma}(\gamma_i)| \leq 2^{-M-1+i}, \quad i = 1, \dots, M-3, \quad 2^{-1} \leq |\hat{\sigma}(\gamma_{M-2})|$$

Since $\hat{\sigma}$ is continuous and (B1) and (B2), such $\gamma_1, \dots, \gamma_{M-3} \in \mathbb{F}$ exists with

$$-M-2+i \leq \gamma_i \leq \frac{-M-1+i}{2}, \quad i = 1, \dots, M-3, \quad 0 < \gamma_{M-2} \leq 1.$$

By (B1), we have

$$2^{-2M-4+2i} \leq |\hat{\sigma}'(\gamma_i)| \leq 2^{\frac{-M-1+i}{2}}, \quad i = 1, \dots, M-3, \quad |\hat{\sigma}'(\gamma_{M-2})| \leq 1.$$

Note that

$$|\gamma_i - \gamma_j| \leq 2^E + 1 \leq 2^{\epsilon_{\max}}, \quad i, j \in \{0\} \cup [M-2].$$

By (B2) and (B3), we have

$$k_i = \lfloor -\frac{M}{2} - 3/2 + i/2 \rfloor_{\mathbb{Z}}, \quad i = 1, \dots, M-3, \quad k_{M-2} = -1$$

and

$$\begin{aligned} |\hat{\sigma}'(\gamma_i)| &\leq 2^{(-M-1+i)/2} \leq 2^{-k_i-M-2+i} \leq 2^{-k_i} |\hat{\sigma}(\gamma_i)|, \quad i = 1, \dots, M-4 \\ |\hat{\sigma}'(\gamma_{M-3})| &\leq 2^{-2} \leq 2^{-k_{M-3}-5} \leq 2^{-k_{M-3}} |\hat{\sigma}(\gamma_{M-3})|, \\ |\hat{\sigma}'(\gamma_{M-2})| &\leq 1 \leq 2^{-k_{M-2}-1} \leq 2^{-k_{M-2}} |\hat{\sigma}(\gamma_{M-2})|. \end{aligned}$$

Then we have

$$\begin{aligned} &\bigcup_{i=1}^{n_0} [\epsilon_{\hat{\sigma}(\gamma_i)} + \epsilon_{\min}, \epsilon_{\max} - \kappa + \min(\epsilon_{\hat{\sigma}(\gamma_i)}, k_i)]_{\mathbb{Z}} \supset \\ &[\epsilon_{\min} - M, \epsilon_{\max} - 2M]_{\mathbb{Z}} \cup \dots \cup [\epsilon_{\min} - 4, \epsilon_{\max} - M - 4]_{\mathbb{Z}} \cup [\epsilon_{\min} - 1, \epsilon_{\max} - M]_{\mathbb{Z}} = [\epsilon_{\min} - M, \epsilon_{\max} - M]_{\mathbb{Z}}. \end{aligned}$$

Condition 2 :

Suppose there is no $\delta_1 \in [0, 2]_{\mathbb{F}}$ such that $\lceil \hat{\sigma}' \rceil(\delta_1) = 2^{-2}$.

Let $A := \{x \in [0, 2]_{\mathbb{F}} : \lceil \hat{\sigma}' \rceil(y) < 2^{-2} \text{ for } y \in [0, 2]_{\mathbb{F}}\}$. We claim that $A \neq \emptyset, [0, 2]_{\mathbb{F}}$.

Suppose $A = \emptyset$. Pick $\xi \in [0, 2] \setminus [0, 2]_{\mathbb{F}}$ such that $\hat{\sigma}'(\xi) = 2^{-2}$. Then there exist $\theta \in [0, 2]_{\mathbb{F}}$ such that $\theta < \xi < \theta^+$ with $\lceil \hat{\sigma}' \rceil(\theta), \lceil \hat{\sigma}' \rceil(\theta^+) > 2^{-2}$. Then we have $\hat{\sigma}'(\theta), \hat{\sigma}'(\theta^+) > 2^{-2} + 2^{-M-3}$. By mean value theorem, there exist $t_1, t_2 \in (\theta, \theta^+)$ such that

$$\hat{\sigma}'(\xi) - \hat{\sigma}'(\theta) = \hat{\sigma}''(t_1)(\xi - \theta), \quad \hat{\sigma}'(\theta^+) - \hat{\sigma}'(\xi) = \hat{\sigma}''(t_2)(\theta^+ - \xi).$$

Since $(\xi - \theta) + (\theta^+ - \xi) \leq 2^{-M}$, we have $(\xi - \theta) \leq 2^{-M-1}$ or $(\theta^+ - \xi) \leq 2^{-M-1}$. Since

$$|\hat{\sigma}''(t_1)| > \frac{2^{-M-3}}{\xi - \theta}, \quad |\hat{\sigma}''(t_2)| > \frac{2^{-M-3}}{\theta^+ - \xi},$$

we have $|\hat{\sigma}''(t_1)| > 2^{-2}$ or $|\hat{\sigma}''(t_2)| > 2^{-2}$ which is contradiction to (B4). Hence $A \neq \emptyset$.

Now suppose $A = [0, 2]_{\mathbb{F}}$. Pick $\xi \in [0, 2] \setminus [0, 2]_{\mathbb{F}}$ such that $\hat{\sigma}'(\xi) = 2^{-2}$. Then there exist $\theta \in [0, 2]_{\mathbb{F}}$ such that $\theta < \xi < \theta^+$ with $\lceil \hat{\sigma}' \rceil(\theta), \lceil \hat{\sigma}' \rceil(\theta^+) < 2^{-2}$. Then we have $\hat{\sigma}'(\theta), \hat{\sigma}'(\theta^+) > 2^{-2} - 2^{-M-4}$. By mean value theorem, there exist $t_1, t_2 \in (\theta, \theta^+)$ such that

$$\hat{\sigma}'(\xi) - \hat{\sigma}'(\theta) = \hat{\sigma}''(t_1)(\xi - \theta), \quad \hat{\sigma}'(\theta^+) - \hat{\sigma}'(\xi) = \hat{\sigma}''(t_2)(\theta^+ - \xi).$$

Since $(\xi - \theta) + (\theta^+ - \xi) \leq 2^{-M}$, we have $(\xi - \theta) \leq 2^{-M-1}$ or $(\theta^+ - \xi) \leq 2^{-M-1}$. Since

$$|\hat{\sigma}''(t_1)| > \frac{2^{-M-4}}{\xi - \theta}, \quad |\hat{\sigma}''(t_2)| > \frac{2^{-M-4}}{\theta^+ - \xi},$$

we have $|\hat{\sigma}''(t_1)| > 2^{-3}$ or $|\hat{\sigma}''(t_2)| > 2^{-3}$ which is contradiction to (B4). Hence $A \neq [0, 2]_{\mathbb{F}}$.

Let $\xi_* := \max A < 2$. Then by (B5), we have $\lceil \hat{\sigma}' \rceil(\xi_*^+) > 2^{-2}$ and $\hat{\sigma}'(\xi_*^+) > 2^{-2} + 2^{-M-3}$. Let $M_* := \max_{x \in (0,2)} |\hat{\sigma}''(x)|$. By (B4), $M_* \leq \frac{1}{10}$ and

$$\hat{\sigma}'(\xi_*) \geq \hat{\sigma}'(\xi_*^+) - M_*(\xi_*^+ - \xi_*) > 2^{-2} + \frac{1}{40},$$

leading to $\lceil \hat{\sigma}' \rceil(\xi_*) \geq 2^{-2}$ which is a contradiction.

Therefore, we have $\delta_1 \in \mathbb{F}$ such that $\lceil \hat{\sigma}' \rceil(\delta_1) = 2^{-2}$.

By (B2) and (B3), $\frac{1}{2} \leq \hat{\sigma}(\delta_1) \leq 2$. Let $\delta_0 = -2^{E+2}$ and $\eta = \epsilon_{\max} - 1$. Since $-2^{2^{E-1}-1} \leq -2^{E+2}$ for $E \geq 4$, we have $\delta_0 \in \mathbb{F}$. By (B1), $0 \leq \hat{\sigma}'(\delta_0) \leq 2^{-2^{E+2}}$ and $\lceil \hat{\sigma}' \rceil(\delta_0) \leq 2^{-2^{E+2}}$. Then we have

$$\begin{aligned} |\delta_0|, |\delta_1|, |\delta_0 - \delta_1| &\leq 2^{E+2} + 2 < 2^{\epsilon_{\max}}, \\ |\lceil \hat{\sigma} \rceil(\delta_0)|, |\lceil \hat{\sigma} \rceil(\delta_1)|, |\lceil \hat{\sigma} \rceil(\delta_0) - \lceil \hat{\sigma} \rceil(\delta_1)| &\leq 3 < 2^{\epsilon_{\max}} \\ -2^{-M-1} \lceil \hat{\sigma}' \rceil(\delta_1) &= -2^{-M-3} < 0 \leq 2 \lceil \hat{\sigma}' \rceil(\delta_0) \leq 2^{-2} = \lceil \hat{\sigma}' \rceil(\delta_1), \\ \max(|\lceil \hat{\sigma} \rceil(\delta_0)|, |\lceil \hat{\sigma} \rceil(\delta_1)|) &\leq 2 \leq 2^{\epsilon_{\max} - \eta}. \end{aligned}$$

Condition 3 :

Take $\zeta = 2^{\epsilon_{\max}-1}$. Now we verify the second item in Condition 3. By (B3), we have $\hat{\sigma}'(0) \geq \frac{1}{4}$ and $\lceil \hat{\sigma}' \rceil(0) \geq \frac{1}{4}$. By (B2) and (B6), we have $\hat{\sigma}([-2\zeta, 2\zeta]) \in [-2, 2] \subset [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. $\square$

Lemma C.9. Suppose continuous function $\hat{\sigma} : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following with $\frac{1}{2} \leq c_1 \leq 1 \leq c_2 \leq \frac{3}{2}$:

- (C1) $\hat{\sigma}(0) = 0$.
- (C2) $8^x \leq |\hat{\sigma}'(x)| \leq 2^x$ for $x \leq -2$.
- (C3) $c_1 x \leq \hat{\sigma}(x), \hat{\sigma}'(x) \leq c_2 x$ for $0 \leq x \leq 1$.
- (C4) $\hat{\sigma}(x)$ is increasing and $\hat{\sigma}'(0) = 1$.
- (C5) $|\hat{\sigma}(x)| \leq 1$ for $x \in \mathbb{R}$.

Then $\lceil \hat{\sigma} \rceil = \sigma$ satisfies Condition 1 with $\kappa = M - 2$, satisfies Condition 2 with $\eta = \epsilon_{\max} - 1$ and satisfies Condition 3 with $\zeta = 2^{\epsilon_{\max}-1}$.

Proof. **Condition 1 :**

Let $\nu = 3, \gamma_1 = 2^{-M-1}, \gamma_2 = 2^{-3}, \gamma_3 = 1, \kappa = M - 2$.

Then we have $k_1 = -M - 3, k_2 = -5, k_3 = -2$ since

$$|\hat{\sigma}'(\gamma_i)| \leq \frac{3}{2} \leq 2^{-k_i} c_1 \gamma_i \leq 2^{-k_i} |\hat{\sigma}(\gamma_i)|, \quad k_i = \epsilon_{\gamma_i} - 2.$$

Since $\epsilon_{\min} \leq -M - 1$, we have

$$\bigcup_{i=1}^{n_0} [\epsilon_{\hat{\sigma}(\gamma_i)} + \epsilon_{\min}, \epsilon_{\max} - \kappa + \min(\epsilon_{\hat{\sigma}(\gamma_i)}, k_i)]_{\mathbb{Z}} \supset$$

$$\begin{aligned} & [\epsilon_{\min} - M, \epsilon_{\max} - 2M - 1]_{\mathbb{Z}} \cup [\epsilon_{\min} - 2, \epsilon_{\max} - M - 3]_{\mathbb{Z}} \cup [\epsilon_{\min} + 1, \epsilon_{\max} - M]_{\mathbb{Z}} \\ &= [\epsilon_{\min} - M, \epsilon_{\max} - M]_{\mathbb{Z}}. \end{aligned}$$

Condition 2 :

By (C4), we have $\hat{\sigma}'(\delta_1) = 1$ for $\delta_1 = 0$. Let $\delta_0 = -2^{E+2}$ and $\eta = \epsilon_{\max} - 1$. Similar to above case, $\delta_0 \in \mathbb{F}$. By (C2), $\hat{\sigma}'(\delta_0) \leq 2^{-2^{E+2}}$ and $\lceil \hat{\sigma}' \rceil(\delta_0) \leq 2^{-2^{E+2}} \leq 2^{-16M}$. Then we have

$$\begin{aligned} & |\delta_0|, |\delta_1|, |\delta_0 - \delta_1| \leq 4\epsilon_{\max} < 2^{\epsilon_{\max}}, \\ & |\lceil \hat{\sigma} \rceil(\delta_0)|, |\lceil \hat{\sigma} \rceil(\delta_1)|, |\lceil \hat{\sigma} \rceil(\delta_0) - \lceil \hat{\sigma} \rceil(\delta_1)| \leq 1 < 2^{\epsilon_{\max}} \\ & -2^{-M-1} \lceil \hat{\sigma}' \rceil(\delta_1) = -2^{-M-1} \leq 2 \lceil \hat{\sigma}' \rceil(\delta_0) \leq 1 = \lceil \hat{\sigma}' \rceil(\delta_1), \\ & \max(|\lceil \hat{\sigma} \rceil(\delta_0)|, |\lceil \hat{\sigma} \rceil(\delta_1)|) = 1 \leq 2^{\epsilon_{\max}-1}. \end{aligned}$$

Condition 3 :

Take $\zeta = 2^{\epsilon_{\max}-1}$. Now we verify the second item in Condition 3. By (C4), we have $\hat{\sigma}'(0) = 1$ and $\lceil \hat{\sigma}' \rceil(0) = 1$. By (C5), we have $\hat{\sigma}([-2\zeta, 2\zeta]) \in [-2, 2] \subset [-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]$. □

C.5. Proof of Lemma C.2

Since conditions (A1)-(A4) implies Conditions 1-3 by Lemma C.7, we need to show that $\hat{\sigma}$ satisfies (A1)-(A4).

For ReLU, ELU, GELU, Swish, ELU, one can verify (A1)-(A3) and the first condition of (A4) by referring Table 3. Also note that $\mathcal{X} = [-\zeta, \zeta]_{\mathbb{F}}$ is $\lceil \hat{\sigma} \rceil$ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{F}}$ by Lemma 3.10 in (Hwang et al., 2025b). where

$$2^{\epsilon_{\max}-2} \leq \zeta = 2^{\epsilon_{\max} - \lceil \log_2(c_2) \rceil - 1} \leq 2^{\epsilon_{\max}-1}.$$

Hence we only need to show the second of (A4), i.e. $|\hat{\sigma}'(x)| \leq 2^x$ for $x \leq -10$.

For $\sigma = \text{ReLU}$, it is trivial since $\sigma'(x) = 0$ for $x \leq 0$.

For $\sigma = \text{ELU} = \begin{cases} x & \text{if } x > 0 \\ e^x - 1 & \text{if } x \leq 0 \end{cases}$, we have $\sigma'(x) = e^x \leq 2^x$ for $x \leq 0$.

For $\sigma = \text{GELU} = x\Phi(x)$ where $\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$ is the standard normal cumulative distribution function. Then we have $\sigma'(x) = \Phi(x) + x\phi(x)$ where $\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$. Note that for $a > 0$,

$$\int_a^{\infty} \phi(t) dt = \int_a^{\infty} \frac{-\phi'(t)}{t} dt = \left[-\frac{\phi(t)}{t} \right]_a^{\infty} - \int_a^{\infty} \frac{\phi'(t)}{t^2} dt \leq \frac{\phi(a)}{a}$$

where we use $\phi'(x) = -x\phi(x)$. Hence for $x < 0$,

$$\Phi(x) = \int_{-\infty}^x \phi(t) dt = \int_{-x}^{\infty} \phi(t) dt \leq \frac{\phi(-x)}{-x}.$$

Therefore, for $x \leq -1$,

$$\sigma'(x) = \Phi(x) + x\phi(x) \leq \frac{\phi(x)}{-x} + x\phi(x) = \phi(x) \left(x - \frac{1}{x} \right) \leq \phi(x) \left(\frac{x^2 - 1}{x} \right) \leq 0,$$

and we have $|\sigma'(x)| \leq x\phi(x)$ since $\Phi(x) > 0$. Hence we need to show $|\sigma'(x)| \leq x\phi(x) \leq 2^x$ for $x \leq -3$. It suffices to show :

$$a\phi(a) = a \frac{1}{\sqrt{2\pi}} e^{-a^2/2} \leq 2^{-a}, \quad a \geq 3.$$

Since both are positive, take natural logs and we have

$$\ln a - \frac{1}{2} \ln 2\pi - \frac{a^2}{2} \leq -a \ln 2.$$

Define $F(\cdot)$ as

$$F(a) = \frac{a^2}{2} - a \ln 2 - \ln a + \frac{1}{2} \ln 2\pi,$$

and we need to show $F(a) \geq 0, a \geq 3$. Note that $F(3) \approx 2.24 > 0$. By taking derivative,

$$F'(a) = a - \ln 2 - \frac{1}{a} > 0, \quad a \geq 3.$$

Since $F(3) > 0$ and $F(a)$ is increasing for $a \geq 3$, we have $F(a) > 0$ for $a > 3$ and $|\sigma'(x)| \leq 2^x$ for $x \leq -3$.

For $\sigma(x) = \text{Swish}(x) = xs(x)$, where $s(x) = \frac{1}{1+e^{-x}}$ is Sigmoid. We have $\sigma'(x) = s(x) + xs'(x) = s(x) + xs(x)(1 - s(x))$. For $x \leq -3$, we have $s(x) = \frac{1}{1+e^{-x}} \geq \frac{1}{1+e^{-3}} > \frac{1}{2}$,

$$\sigma'(x) = s(x)(1 + x(1 - s(x))) < s(x) \left(1 + (-3)\left(\frac{1}{2}\right)\right) = s(x)\left(-\frac{1}{2}\right) < 0.$$

Since $\sigma'(x) < 0$ for $x \leq -6$, we have

$$|\sigma'(x)| = |s(x) + xs(x)(1 - s(x))| \leq |xs(x)(1 - s(x))| \leq \left|\frac{x}{1+e^{-x}}\right| \leq |xe^x| \leq 2^x.$$

for $x \leq -10$.

C.6. Proof of Lemma C.3

Since conditions (B1)-(B6) implies Conditions 1-3 by Lemma C.8, we need to show that $\hat{\sigma}$ satisfies (B1)-(B6).

One can verify (B2)-(B6) by referring Table 4.

By (B1), there exists a separating point $\theta_1 \leq -4$. By (B2), there exist a separating point $\theta_2 \in [0, 2]_{\mathbb{R}}$. By Lemma 3.7 in (Hwang et al., 2025b), $\mathcal{X} = [-\zeta, \zeta]_{\mathbb{R}}$ is $[\hat{\sigma}]$ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{R}}$, where $\zeta \leq \Omega$.

Hence we only need to show the second of (B1), i.e. $4^x \leq |\hat{\sigma}(x)|, |\hat{\sigma}'(x)| \leq 2^x$ for $x \leq -2$. Since $1 + e^t \leq 4^t$ for $t \geq 2$, we have

$$4^x \leq \hat{\sigma}(x) = \frac{1}{1+e^{-x}} \leq e^x \leq 2^x, \quad x \geq -2.$$

Since $\hat{\sigma}'(x) = \hat{\sigma}(x)(1 - \hat{\sigma}(x))$, we have

$$\hat{\sigma}(x)(1 - \hat{\sigma}(-2)) \leq \hat{\sigma}(x)(1 - \hat{\sigma}(x)) \leq \hat{\sigma}(x) \leq 2^x, \quad x \geq -2.$$

We need to show $\hat{\sigma}(x)(1 - \hat{\sigma}(-2)) \geq \hat{\sigma}(x)(0.8) \geq 4^x$. Since $\hat{\sigma}(-2)(0.8) \approx 0.105 \geq 4^{-2}$ and 4^x is decreasing faster as $x \rightarrow -\infty$, we get the desired result.

C.7. Proof of Lemma C.4

Since conditions (C1)-(C5) implies Conditions 1-3 by Lemma C.9, we need to show that $\hat{\sigma}$ satisfies (C1)-(C5).

By (C2), there exists a separating point $\theta_1 \leq -4$. By (C3), there exist a separating point $\theta_2 \in [0, 1]_{\mathbb{R}}$.

By Lemma 3.7 in (Hwang et al., 2025b), $\mathcal{X} = [-\zeta, \zeta]_{\mathbb{R}}$ is $[\hat{\sigma}]$ -distinguishable with range $[-2^{\epsilon_{\max}}, 2^{\epsilon_{\max}}]_{\mathbb{R}}$, where

$$\zeta \leq \Omega.$$

One can verify (C1)-(C5) since $\hat{\sigma}(0) = 0, \hat{\sigma}'(0) = 1, 0.76x \leq \hat{\sigma}(x) \leq x$ for $0 \leq x \leq 1$, $\hat{\sigma}(x)$ is increasing and $|\hat{\sigma}(x)| \leq 1$ for $x \in \mathbb{R}$. We only need to show $8^x \leq |\hat{\sigma}'(x)| \leq 2^x$ for $x \leq -2$ and we have

$$8^x \leq e^{2x} \leq \frac{4e^{2x}}{4} \leq \hat{\sigma}'(x) = \frac{4e^{2x}}{(1+e^{2x})^2} \leq 4e^{2x} \leq 2^x.$$

$\hat{\sigma}(x)$	γ_0	(c_1, c_2)	$\hat{\sigma}'((0, \infty))$	$\hat{\sigma}''((a, b))$	$\hat{\sigma}'([a, b])$	$\hat{\sigma}((-\infty, 0])$	$f(x) = \frac{\hat{\sigma}'(x)}{2^{-x}}, f(-\infty, -3)$
ReLU	0	(1,1)	{1}	{0}	{1}	{0}	{0}
ELU	0	(1,1)	{1}	{0}	{1}	$\subset [-1, 0]$	$\subset (-0.08, 0)$
GELU	0	(0.5,1)	$\subset [0.5, 1.13]$	$\subset (0.24, 0.62)$	$\supset (0.85, 1.08)$	$\subset (-0.18, 0]$	$\subset (-0.002, 0)$
Swish	0	(0.5,1)	$\subset (0.5, 1.10)$	$\subset (0.05, 0.31)$	$\supset (0.93, 1.09)$	$\subset ((-0.28, 0))$	$\subset (-0.03, 0)$

Table 3. Properties of various real activations functions. Note that (a, b) is $(\frac{1}{2}, 2)$ for $\hat{\sigma} = \text{ReLU}, \text{ELU}$, $(\frac{1}{2}, 1)$ for $\hat{\sigma} = \text{GELU}$ and $(1, 2)$ for $\hat{\sigma} = \text{Swish}$.

$\hat{\sigma}(x)$	γ_0	(c_1, c_2)	$\hat{\sigma}'(0)$	$\hat{\sigma}''(\mathbb{R})$	$\hat{\sigma}'([0, 2])$	$\hat{\sigma}((-\infty, 0])$
Sigmoid	0	(0.5,1)	$\frac{1}{4}$	$\subset (-0.10, 0)$	$\subset (0.1, 0.25]$	$(0, 0.5]$
$\hat{\sigma}(x)$	γ_0	(c_1, c_2)	$\hat{\sigma}'(0)$	$\hat{\sigma}''(\mathbb{R})$	$\hat{\sigma}'([0, 2])$	$\hat{\sigma}((-\infty, 0])$
tanh	0	(0.7616,1)	$\frac{1}{4}$	$\subset (-0.10, 0)$	$\subset (0.1, 0.25]$	$(0, 0.5]$

Table 4. Properties of floating-point format for verifying the conditions.